

Metagame, Time Use, and Player Development in Online Chess: A Large-Scale Study of Lichess Games

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Abstract

Metagame change, time management, and player development are central but often separately studied dimensions of online chess behavior. Using 24 months of public Lichess data from January 2024 through December 2025, this study builds a reproducible large-scale pipeline that samples rated games and derives linked datasets for metagame composition, opening repertoire, time use, and longitudinal player outcomes. The analysis is observational and non-causal, combining descriptive modeling, multinomial opening models, player fixed-effects progression models, and clustering-based behavioral profiling with forward-in-time evaluation. Results indicate that the opening environment gradually spreads across a wider set of choices over time, although the extent of that spread differs by time control. Opening preferences also shift with skill, as stronger players devote a smaller share of games to the most common named openings. Patterns of time use vary by format: Higher-rated players move faster in blitz, think longer in rapid and classical, and enter low-time situations more often across all formats. In models of within-player change, later rating gains are associated more consistently with richer and less static opening repertoires and with more complex games than with sheer game volume. Clustering results likewise suggest that chess behavior is better understood as a set of recurring play states than as a small number of fixed player types. Together, these findings suggest that online chess improvement is associated less with sheer volume of play than with the quality and adaptability of engagement, while time-use patterns separately show that players manage pace and pressure in format-specific ways. More broadly, the results show how large-scale game logs can be used to study adaptation and skill development in online chess environments.

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1 Introduction

Competitive games have become an important source of large-scale behavioral data, enabling researchers to study strategic decision-making, skill development, and population-level trends using observational logs at scales rarely achievable in controlled studies (Edelmann et al. 2020, Watts 2014). Chess is one of the most-played competitive games in the world, and the relatively recent growth of online platforms has created datasets with millions to billions of games played under standardized rules and well-defined competitive contexts (e.g., time controls, ratings, game outcomes) (Watts 2014, Lichess.org 2026b). One of the most widely used sources is the Lichess Open Database, which provides public monthly exports of rated games played on lichess.org in PGN format (Lichess.org 2026b, Lichess contributors 2026). These exports make it possible to study the ways strategy and performance vary across skill levels and time controls in a modern, online environment.

This study uses Lichess game data to investigate contemporary metagame behavior and player development. Here, ‘metagame’ refers to the population-level distribution of strategic choices and styles observed in practice – such as opening families, risk preferences, and game outcomes – within a defined time window and player population. In addition to describing how strategic behavior differs by rating and time control, we attempt to connect these strategic signatures to measures of player progression over time. Because online chess is a repeatedly played competitive setting with a mature rating infrastructure, it offers a natural testbed for quantifying skill and comparing behavior across a large player population (Lichess.org 2026a, Glickman 2022).

A central component of modern chess analytics is the rating system used to summarize player strength. The Elo rating system, originally developed for chess, models player per-

formance probabilistically, and remains a foundational reference point for interpreting skill differences and expected outcomes in competitive settings (Elo 1978, International Chess Federation 2025). Lichess uses a modified version of this system called ‘Glicko-2’, which, in addition to rating, tracks uncertainty via rating deviation and volatility (Lichess.org 2026a, Glickman 2022). Skill ratings are especially useful for observational studies like this one because they provide a consistent scale by which to stratify players and construct longitudinal measures of improvement (Glickman 2022).

Overall, this project brings together data science, computational social science, and strategic-behavior research by treating online chess as a real-world environment in which strategic choices and performance outcomes can be measured at scale and analyzed using statistical learning methods (Edelmann et al. 2020, Watts 2014).

1.1 Background

A growing research literature uses large online chess datasets to study human performance, decision-making, and strategic patterns. For example, Chowdhary et al. use an online chess dataset of more than 100 million games to quantify performance dynamics and career-like trajectories, illustrating how large game archives support population-level behavioral measurement across skill levels (Chowdhary et al. 2023).

One important line of work focuses on time usage and decision quality under time constraints. Online chess platforms record both move sequences and clock information, which allows researchers to study how players allocate thinking time and how time pressure shapes decisions. Russek et al. analyze millions of Lichess games to relate move times to the estimated value of additional computation, and show that time allocation patterns vary systematically with playing strength and time control (Russek et al. 2025). These findings

support the idea that online chess is a rich setting for studying bounded rationality and performance under constraints, especially when comparing bullet, blitz, rapid, and classical formats ([Russek et al. 2025](#)).

Chess opening choice is also an active research topic. Because openings are discrete, labelable strategic choices that occur early in every game, they are well suited to population-level analysis. De Marzo and Servedio use online chess community data to quantify opening similarity and complexity, including representing openings in network form and studying how opening-related structure can be used to forecast future opening adoption ([De Marzo & Servedio 2023](#)). This motivates analyzing metagame development in terms of shifting opening popularity and success rates across time, player skill, and time controls ([De Marzo & Servedio 2023](#), [Chowdhary et al. 2023](#)).

Finally, there is an emerging interest in identifying player styles or archetypes from game logs. While traditional chess analysis often focuses on engine evaluation of positions, a complementary approach is to extract behavioral features from move sequences and outcomes (e.g., game length, opening selection, and other style proxies) and use statistical learning to characterize player types. For instance, McIlroy-Young et al. demonstrate that models trained on move sequences can capture individual-specific signatures (‘style’) in player decisions, supporting the notion that player behavior is meaningfully distinguishable from game records alone ([McIlroy-Young et al. 2021](#)). This creates a bridge between strategic analysis (what players choose to do) and longitudinal outcomes (how players improve or persist over time).

Collectively, existing work suggests that large-scale online chess datasets support three relevant kinds of analysis: (1) descriptive metagame analysis of strategic choices across populations; (2) modeling the way constraints (such as time pressure) affect behavior and

outcomes; and (3) data-driven modeling of player growth and characterization of potential player types (Chowdhary et al. 2023, Russek et al. 2025, De Marzo & Servedio 2023, McIlroy-Young et al. 2021). Here, we build on those themes by using Lichess data to analyze contemporary opening and style patterns and relate them to measures of player performance and development over time.

2 The Data

2.1 Source

Lichess is the largest open-source chess server in the world and releases free monthly exports of its game data under the Creative Commons CC0 license (Lichess.org 2026b). The site logs game data while a game is being played, and allows users to retroactively analyze both the games they have played and those of others on the platform. Although Lichess offers a powerful API, we elected to download data directly from their website in `.pgn.zst` format. The PGN (portable game notation) format is a plaintext format for chess games, containing moves, metadata headers (e.g., players, date, and event), and optional annotation. The `.zst` file extension denotes Zstandard compression, a modern compression format commonly used for its speed and efficiency. In practice, Zstandard substantially reduces the storage required for large PGN archives.

For the purposes of this study, we downloaded 24 months of Lichess game data spanning from January 2024 to December 2025, amounting to approximately 677 gigabytes of raw data when compressed. This volume – nearly 700 gigabytes compressed, or roughly 3–4 terabytes uncompressed – was too large to process conveniently on local hardware alone. We therefore relied on a hybrid R/Python pipeline that downloads the data, determin-

istically samples and parses it into partitioned parquet files, and then constructs smaller analysis-ready datasets from those outputs.

The download and parsing section of the pipeline performs the following:

- Reads a config file `pipeline.yml` that controls the scope and parameters of the pipeline’s operation.
- Downloads monthly Lichess ‘standard rated’ PGN archives for each month specified in the YAML.
- Decompresses each Zstandard file as a stream, scanning games one-by-one to prevent memory bottlenecks.
- Parses game headers first to determine whether a game meets the sampling and filtering criteria before doing more expensive parsing work.
 - Extracts core metadata (players, ratings, time control, result, termination, opening, timestamps).
 - Buckets each game into blitz/rapid/classical using a Lichess-style estimated-duration rule based on $\text{initial} + 40 \times \text{increment seconds}$.
- Applies filtering rules from the YAML, which in this case are:
 - Include only blitz/rapid/classical time controls.
 - Include only games from players in the 800–2600 rating range.
 - Drop unwanted terminations (aborted games).
- Uses deterministic hash-based sampling to create reproducible subsets without needing to store the whole population:
 - A meta sample keyed by `game_id` for population/metagame summaries, at a rate of 5/500 games.

- A player cohort keyed by `username` for longitudinal/player tracking at a rate of 3/500 players.
- A moves sample keyed by `game_id` for clock/time-per-move data at a rate of 5/10000 moves.
- Deterministically anonymizes usernames based on a stable, salted hash.
 - Cohort selection is computed on the real username before anonymization, so it remains reproducible.
- Writes month-partitioned parquet outputs for game-level data.

The dataset-building portion of the pipeline takes those outputs and, using DuckDB to read and write parquet files on disk, builds seven datasets from them:

- Player-level longitudinal summary tables from `games_cohort`:
 - `player_month (pm)`: $\text{player} \times \text{month}$, 249,761 observations.
 - `player_month_tc (pm_tc)`: $\text{player} \times \text{month} \times \text{time control}$, 320,869 observations.
- Player-opening ‘repertoire’ table from `games_cohort`:
 - `player_opening_month_tc (pom_tc)`: $\text{player} \times \text{month} \times \text{time control} \times \text{opening}$, 7,167,580 observations.
- Metagame tables from `games_meta`:
 - `meta_opening_month_tc (mom_tc)`: $\text{openings} \times \text{month} \times \text{time control}$, 130,672 observations.
 - `meta_month_tc (mm_tc)`: $\text{month} \times \text{time control}$, 72 observations.
- Time-usage tables from `moves_meta`:

- `player_time_month_tc` (`ptm_tc`): $\text{player} \times \text{month} \times \text{time control}$, 1,038,050 observations.
- `moves_month_tc` (`mvm_tc`): $\text{month} \times \text{time control}$, 72 observations.

The pipeline logs session and version info for both R and Python and records checksums for downloaded raw files to help with reproducibility.

2.2 Variables

The analysis uses several linked derived datasets rather than a single analytic table. In all cases, the raw unit of data is a rated chess game from the Lichess public database, but the pipeline aggregates games and sometimes moves into analysis-ready summaries. At the player level, `pm` and `pm_tc` summarize monthly outcomes overall and by time control; `pom_tc` captures opening use within player-month-time-control cells; `ptm_tc` summarizes move-time behavior; and `mom_tc` and `mm_tc` provide population-level metagame summaries. Because these tables operate at different units of analysis, the main text introduces them only at the level needed for later modeling. Full variable definitions are reported in Appendix [A](#).

Together, these analytical levels support longitudinal player-performance modeling (`pm`, `pm_tc`), repertoire and style measurement from opening choice (`pom_tc`), metagame analysis via opening adoption and outcomes (`mom_tc`), and time-management analysis using move-time features (`ptm_tc`).¹

¹The remaining tables, `mm_tc` and `mvm_tc`, are summary-level datasets used mainly for aggregate diagnostics and macro-level summaries rather than player-level modeling.

2.3 Cleaning

Most cleaning steps were performed during the pipeline’s parsing stage. Additional standardization and feature construction steps not covered in Section 2.1 include:

- `TimeControl` tag parsed into:
 - `initial` (seconds).
 - `increment` (seconds).
 - `tc_bucket`, based on estimated duration $\text{initial} + 40 \times \text{increment}$ seconds:
 - * < 180 = `bullet`, < 600 = `blitz`, < 1800 = `rapid`, else `classical`.
- Converted PGN result strings into:
 - `winner` (white/black/draw).
 - `score_white` (1/0.5/0).
- Computed upset indicators using ratings:
 - `rating_diff` = `white_rating` - `black_rating`.
 - `abs_rating_diff`
 - `is_upset`
- Computed game length in `moves` from move sequences for use in the derived datasets.

The resulting data were then restructured and aggregated into tidy analysis tables.

3 Exploratory Data Analysis

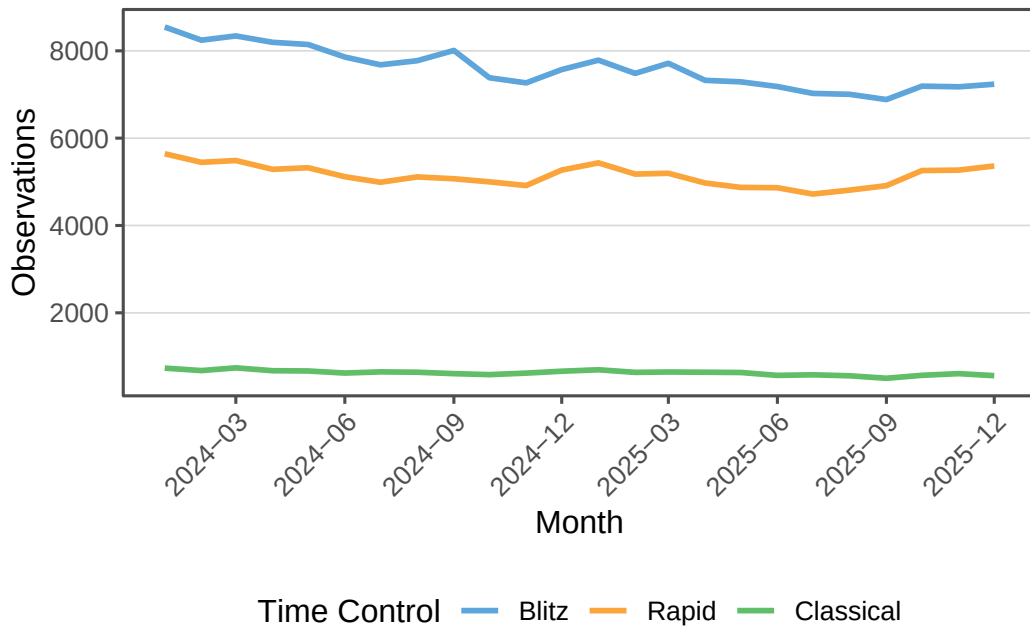
This section summarizes the main descriptive patterns in the sampled data before formal modeling. It first documents support and reliability constraints that affect interpretation,

then turns to broad patterns in opening behavior and time usage. Full supporting diagnostics are reported in Appendix B.

3.1 Coverage and Reliability

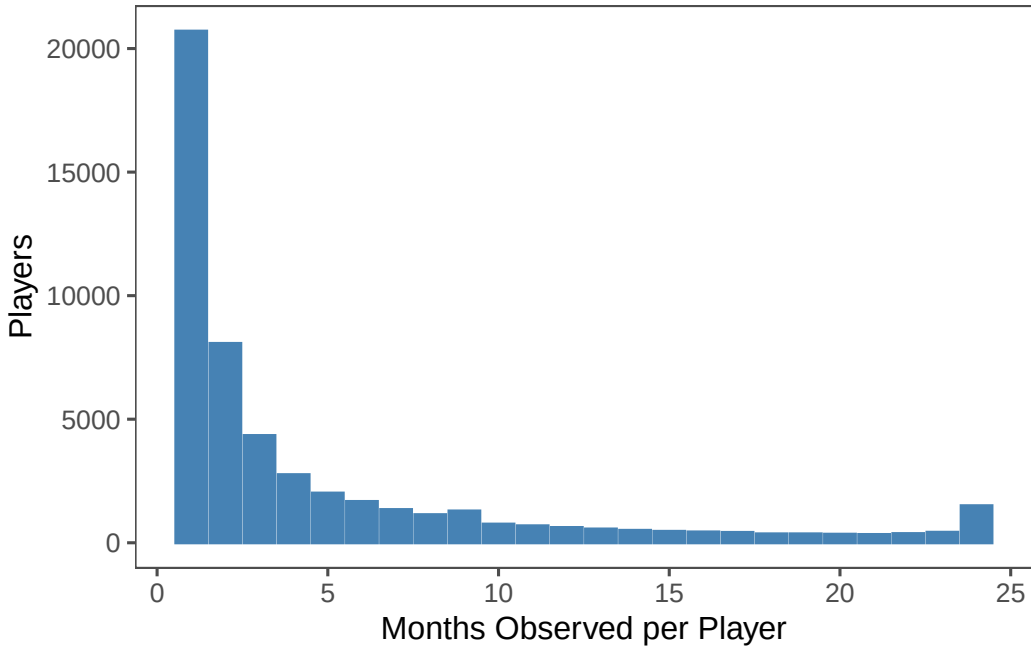
Before turning to observed patterns, it is useful to clarify which parts of the sampled data are more and less strongly supported. Figure 1 shows that the number of observations differs substantially across time controls, with blitz appearing more often than rapid, and both appearing much more often than classical. This is a direct consequence of our parsing method: The pipeline sampled from the underlying game population rather than enforcing balanced coverage by format. As such, the cleaned data preserve the activity structure of the platform itself. That is not a problem in its own right, but it does mean that comparisons across time controls come with unequal precision, especially where classical is concerned.

Figure 1: Coverage of player-month-time-control observations over time by time control.



A second limitation is that the player panel is unbalanced. Figure 2 shows that many players appear for only a few months, while a much smaller subset remain observable across all or most of the two-year window. This pattern is expected in an online hobby setting, but it matters once the analysis shifts from cross-sectional summaries to player-level change over time. Any lagged or forward-looking quantity necessarily relies on re-observing the same player in later months, so longitudinal analyses operate on a smaller and more selective subset than the full sampled panel.

Figure 2: Distribution of months observed per player in the sampled panel.



Overall, the data are informative but unevenly supported. Some parts of the sample are much better represented than others, so short-horizon player outcomes should be interpreted with appropriate caution. Section 4 and Section 5 address these issues more directly.

3.2 Opening Behavior

The opening plots suggest considerably more temporal stability than rapid metagame turnover. At the level of the most common named openings, there is not much evidence of dramatic month-to-month turnover across the two-year sample window; the basic impression is one of gradual change as opposed to rapid metagame cycling. Although the degree of stability differs by time control, the same broad pattern appears across all three formats. Additional time-control-specific plots are reported in Appendix B.

Figure 3: Opening share by rating for most common blitz openings.

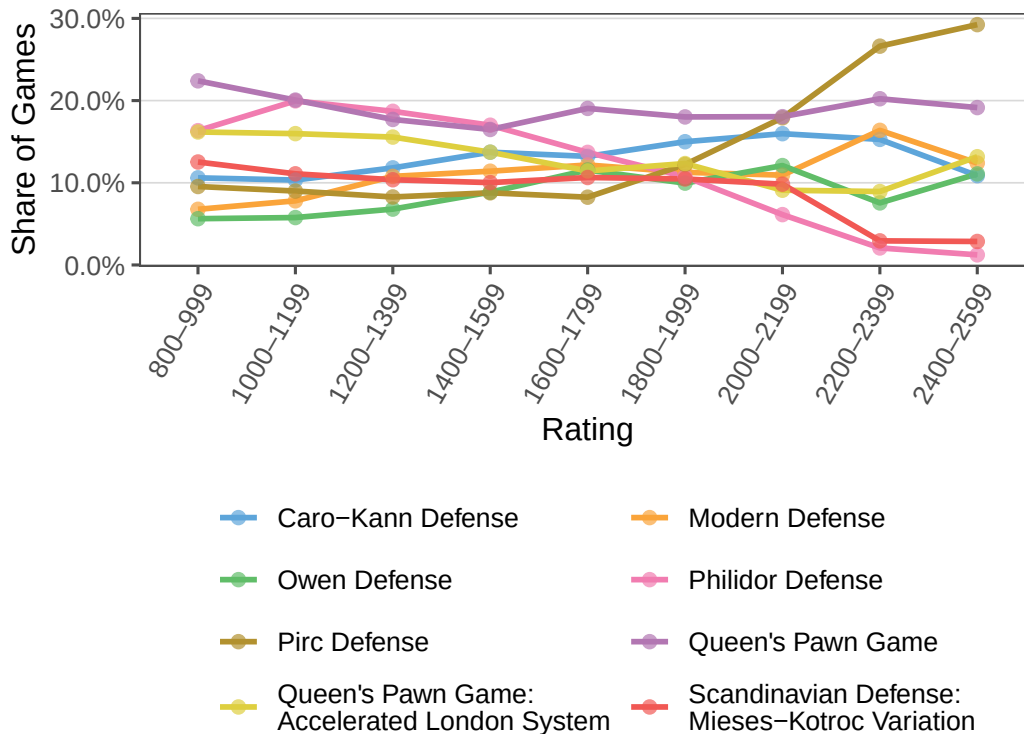


Figure 3 is more informative for interpretation. Although broad temporal drift appears fairly modest, opening share changes more clearly across rating levels. Some openings gain relative share as player rating increases, while others lose it, and those gradients are easier to see than the changes over calendar time. The Pirc Defense and Philidor Defense are

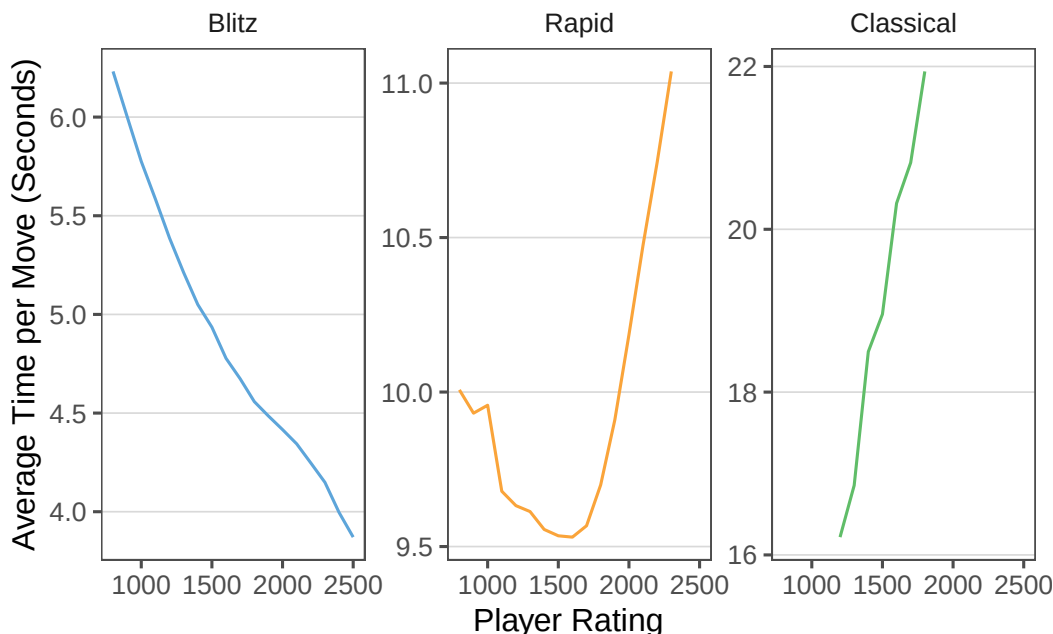
especially good examples. Even where the shifts are not large in absolute terms, the rating structure is clearer than anything suggested by the over-time plots alone.

In summary, the exploratory evidence suggests that opening behavior is better framed as structured choice by rating and format, with gradual long-tail diversification, than as rapid turnover among named openings.

3.3 Time Use

The time-usage plots show clearer structure. Figure 4 shows a clear difference in thinking time across both time controls and skill levels. Blitz and classical show opposite gradients: Stronger players spend less time per move in blitz, but more time per move in classical. Rapid lies somewhere in between. Even without formal modeling, this shows that time control is not just a background element; it fundamentally changes the relationship between rating and pace.

Figure 4: Average time spent per move by player rating and time control.



A useful supporting pattern is that the same broad structure remains visible when time use is broken down by game phase, suggesting that the broad time-use gradients in Figure 4 are not just artifacts of averaging everything together. Appendix B reports the corresponding phase-specific diagnostics. All in all, time management does not look one-dimensional. Average pace changes across formats, and the meaning of ‘using more time’ depends heavily on the time control in question, which makes time usage look like a genuine behavioral outcome rather than a mechanical byproduct of format alone.

4 Methodology

This section describes the analytic preparation, feature construction, and modeling framework used in the study. The analysis is observational and non-causal throughout; models are used for description, association, prediction, and exploratory summarization, *not* causal identification.

Because the project combines several research questions across multiple linked datasets, the methodology proceeds in two stages. First, variables are transformed and aligned so that quantities are comparable across players, months, and time controls. Second, we define a set of modeling tasks aligned with the project goals: metagame dynamics, opening popularity, time use, player progression, and player categorization.

All predictive modeling uses forward-in-time data splits. Training data are drawn from earlier months in the sample window, while validation and test data are drawn from later contiguous months. Splitting is performed at the month level so that observations from a given month are never used to predict themselves or earlier periods. This design reduces temporal leakage and makes held-out evaluation more realistic, especially for progression

targets defined for future rating change. Where relevant, models are also checked across time-control and rating strata to ensure that conclusions are not driven by a single format or skill band.

4.1 Data Transformations

The linked analytic tables described in Section 2 operate at different observational units and therefore require additional alignment before modeling. Transformations were implemented to improve comparability across contexts, reduce instability in sparse cells, and construct interpretable predictors for the later analyses.

4.1.1 Analytic Alignment and Scaling

The project does not rely on a single modeling table. Instead, different tasks use different analytic units, including player-month summaries, player-month-by-time-control summaries, opening-level repertoire tables, metagame month summaries, and move-time summaries. These tables were retained at their native units whenever possible so that each model could be fit on the level most appropriate to its target. As a result, some models operate on aggregate month-level data, while others operate on player-period or opening-period observations.

Rating was sometimes converted from a continuous quantity into bins for descriptive summaries, nonparametric plots, and models that benefited from coarse stratification. This was done primarily to stabilize estimates and improve interpretability in settings where fine-grained rating variation was not necessary. For clustering and other multivariate feature-based analyses, retained numeric inputs were standardized as z-scores so that variables

measured on different scales – e.g., seconds, proportions, counts, rating points – would contribute on a comparable basis.

4.1.2 Weighting, Thresholds, and Support

Many outcomes are rates or shares estimated from cells with unequal support. Some observations are based on large numbers of games or move rows, whereas others are based on very small samples and are correspondingly volatile. For this reason, weighting and support restrictions were used where appropriate.

When player-level quantities needed to be collapsed across time controls, overall summaries were constructed using exposure-based weighting. For example, overall player-month rating was computed as a games-weighted average of time-control-specific average ratings, and overall time-pressure quantities were computed using move-based weighting when derived from move-level summaries. This approach prevents low-volume cells from exerting the same influence as high-volume cells when constructing cross-format summaries.

Minimum-support thresholds were also applied in settings where extremely sparse observations would otherwise dominate. Depending on the task, these thresholds were defined in terms of games per player-month, games per opening cell, or move rows per time-use summary. These filters were not intended to alter the substantive population under study, but rather to limit the influence of cells whose estimates were too unstable to reliably interpret.

4.1.3 Derived Features and Missingness

Several interpretable features were constructed from the player-opening and time-use tables for use in the progression and categorization analyses. These features were intended to

summarize broad aspects of player behavior.

Repertoire breadth was measured using opening diversity, defined as the number of distinct openings used within the relevant period. Repertoire concentration was summarized using the share of games allocated to the player’s most-used opening. Repertoire adjustment was captured in two related ways— an indicator for whether the most-played opening changed from the previous month, and measures of month-to-month repertoire churn based on changes in opening-share vectors and the appearance of newly used openings. Time-use behavior was summarized using variables such as average time spent per move, upper-tail move-time summaries, opening-phase time allocation, and low-time frequency.²

Some time-use variables were structurally missing rather than randomly absent. In particular, endgame time measures are often unavailable because many games do not reach the move-number threshold used to define the endgame phase. These missing values therefore reflect game structure rather than arbitrary loss of data. Models using phase-specific variables either handled these cases explicitly or restricted analysis to observations for which the relevant phase was meaningfully defined.

4.2 Modeling Tasks and Candidate Specifications

Because this paper addresses several distinct questions, candidate models are organized by modeling task rather than abstract model family. Each task corresponds to a substantive question, a natural observational unit, and a set of candidate specifications suited

²Phase-specific time-use variables use move-count intervals rather than engine-defined phases: opening is moves 1–10, middlegame is moves 11–25, and endgame is moves 26 and later. Low-time definition: A move is flagged as “low time” if the remaining clock time after the move satisfies $\text{clock_after} \leq \max(10, 2 \times \text{increment})$ seconds, where increment is the per-move increment of the game’s time control.

to that problem. Formal model comparison and retention are deferred to Section 5. The present section instead defines the role of each modeling task and the types of specifications considered.

4.2.1 Metagame Dynamics

Metagame-dynamics models were used to describe how aggregate ecosystem measures changed across month and time control. These models focus on population-level structure rather than individual players. Depending on the outcome, the primary observational units were either month-by-time-control summaries or opening-by-month-by-time-control summaries.

The main outcomes in this task include concentration and diversity measures, draw rate, upset rate, average game length, and opening share. Predictors primarily include month, time control, and their interactions. Candidate specifications included weighted linear models for macro outcomes and multinomial models for opening composition, with simpler opening-by-opening share models considered as descriptive comparisons.

The motivation for using more than one specification in this task is straightforward. Some metagame quantities, such as concentration or average moves, are scalar month-level summaries and are naturally modeled with weighted regressions. Opening composition, by contrast, is inherently compositional— increases in one opening’s share imply decreases elsewhere. Models that account for this compositional structure were therefore considered when the goal was to describe substitution within the opening ecosystem.

4.2.2 Opening Popularity

Opening-popularity models were used to describe how opening choice varied with player strength and time control. This task overlaps with metagame analysis but is narrower in scope; rather than modeling the ecosystem as a whole, it focuses specifically on opening-adoption patterns.

The primary observational unit for this task is the month-by-time-control-by-rating-bin opening summary. Outcomes are opening shares or opening-use counts, and the key predictors are rating level, time control, month, and rating-by-format interactions. Candidate specifications included pooled multinomial models as the main compositional approach, along with count-based and weighted share models as lower-complexity alternatives.

This task treats opening popularity as a structured choice problem. The multinomial framework is useful because it represents the full set of retained openings jointly, allowing probabilities to coherently shift across categories. Simpler single-opening models remain useful as descriptive checks, but fail to fully represent substitution across openings and as such play a secondary role in this modeling category.

4.2.3 Time Use

Time-use models were used to characterize how time allocation varied across skill and time control. These models treat time management as a behavioral outcome rather than as a byproduct of format alone. Two related targets were emphasized— average time use and low-time frequency.

Depending on the target, the primary observational units were either player-month-by-time-control summaries or month-level time-use summaries. Outcomes include average

time spent per move, upper-tail move-time summaries, phase-specific averages, and the proportion of moves taken in low-time states. The central predictors are player rating, time control, and in some cases their interaction, with month included where temporal adjustment is needed.

Candidate specifications included additive and interaction regressions for mean time-use outcomes, robust or quantile-based alternatives for heavy-tailed variables, and bounded-outcome models for low-time frequency. The reason for considering multiple mean-modeling strategies is that time-use data can be skewed and sensitive to outliers, while low-time frequency is both bounded and concentrated near zero. Different specifications were considered depending on whether the main challenge was nonlinearity, heavy tails, or bounded proportion structure.

4.2.4 Player Progression

Player-progression models were used to relate current-month behavior to later rating movement. These models ask whether within-player changes in activity, game structure, repertoire behavior, and related features are associated with subsequent changes in rating over short horizons.

The baseline observational unit is the player-month table, optionally joined to time-control or repertoire summaries when the required predictors are available. Outcomes are defined as future rating change over one- and three-month horizons, along with an absolute-change outcome used to summarize movement magnitude without direction. Predictors include current rating, game volume, average game length, repertoire breadth, repertoire concentration, repertoire adjustment, and selected time-use features where support permits.

Candidate specifications included regularized linear models, panel-data models, and

tree-based regressions. Regularized models were considered because many player-level predictors are correlated. Panel models were considered because the data are longitudinal and repeated within players. Tree-based models were considered as flexible predictive benchmarks capable of capturing nonlinearities and interactions without extensive manual specification. Because this task is explicitly forward-looking, realistic temporal splitting is especially important; the model must predict later rating movement using only information available at the current month.

4.2.5 Player Categorization

Player-categorization models were used to summarize recurring behavioral profiles in player-month observations. This task is exploratory rather than inferential. The goal is not to estimate latent causal types, but to determine whether broad behavioral configurations can be recovered from performance, repertoire, and activity features.

The natural observational unit is a joined player-month-by-time-control table containing performance measures, repertoire summaries, and other standardized features. Inputs include rating, rating differential, game length, outcome rates, repertoire breadth, repertoire concentration, repertoire adjustment, and selected activity measures. Time-use variables were considered where coverage allowed, but their inclusion depended on support in the merged sample.

Candidate specifications included dimensionality-reduction methods and clustering algorithms. Dimensionality reduction was used to summarize major axes of behavioral variation and to inspect whether lower-dimensional structure was visible in the data. Clustering methods were used to partition the feature space into interpretable behavioral groupings. Because the goal of this task is descriptive summarization, later evaluation emphasizes

balance, stability, and interpretability rather than formal inference.

4.3 Summary

Taken together, these transformations and modeling tasks define the analytic structure of the paper. The transformations align linked datasets, stabilize sparse quantities, and produce interpretable features for later use. The modeling tasks then organize the empirical questions into five domains: ecosystem-level metagame change, opening choice, time management, longitudinal progression, and exploratory behavioral profiling.

5 Modeling

This section compares the candidate models introduced in Section 4.2 and identifies the specifications retained for the remainder of the analysis. The purpose here is not to present substantive findings, but to narrow a broader set of candidate models to a smaller set of primary specifications for each modeling task. Substantive interpretation is deferred to Section 6.

Model retention is based on four considerations: inferential suitability, held-out predictive adequacy, robustness to reasonable specification changes, and interpretability. These criteria do not apply identically to every modeling problem. In some cases, the retained model is primarily inferential; in others, it is better understood as descriptive or predictive. This distinction is especially important for exploratory methods such as clustering, which are useful for summarizing structure in the data but are not treated as a basis for inferential claims.

Because a large number of models were fit during development, the body of this section

focuses on retention decisions and the reasons for them. Detailed predictive comparisons, robustness checks, and supplemental diagnostics are reported in Appendices C, D, E, and F.

5.1 Metagame Dynamics

Metagame dynamics were modeled at two related levels. First, aggregate ecosystem and game-environment outcomes were modeled at the month-by-time-control level. These macro outcomes included opening concentration and diversity, along with draw rate, upset rate, and average moves. Second, opening composition itself was modeled separately by time control using specifications designed for compositional outcomes.

For the macro outcomes, weighted linear models were retained as the primary inferential framework. These specifications provided a direct and interpretable way to estimate time trends and time-control differences while accommodating unequal support across month-by-format cells. Because the macro series are observed over time, inference relied on heteroskedasticity- and autocorrelation-consistent standard errors.

Figure 5: Retained macro metagame model.

$$Y_{c,t} = \beta_0 + \beta_1 t + \beta_2 \mathbf{1}\{c = \text{Classical}\} + \beta_3 \mathbf{1}\{c = \text{Rapid}\} \\ + \beta_4 t \mathbf{1}\{c = \text{Classical}\} + \beta_5 t \mathbf{1}\{c = \text{Rapid}\} + \varepsilon_{c,t}.$$

Here, $Y_{c,t}$ denotes a month-by-time-control metagame outcome, such as HHI concentration, effective number of openings, draw rate, upset rate, or average moves. Blitz is the reference time-control category. Inference used Newey–West heteroskedasticity- and autocorrelation-consistent standard errors.

Opening composition required a different retained model because opening shares are

inherently compositional. Candidate models included weighted fixed-effects share regressions, GAM-based share regressions, and multinomial specifications fit separately by time control. The multinomial models were retained as the primary inferential basis for compositional change because they represent the full opening set jointly and therefore respect the substitution structure of opening shares. By contrast, the share-based alternatives model openings one at a time and are better treated as descriptive comparisons than as primary inferential models.

Figure 6: Retained multinomial metagame-composition model, fit separately by time control.

$$\Pr(Y_t = k) = \frac{\exp(\eta_{k,t})}{\sum_{j=1}^K \exp(\eta_{j,t})}, \quad k = 1, \dots, K,$$

$$\eta_{k,t} = \alpha_k + \beta_k t, \quad k = 1, \dots, K-1,$$

$$\eta_{K,t} = 0.$$

Here, Y_t denotes the opening category at month t , and k indexes the retained opening categories.

The final category, K , is the reference category, so category-specific effects are interpreted relative to that baseline.

Robustness checks further supported this division of labor. Changes to the retained opening set altered predictive difficulty in the expected way, but did not overturn the main directional patterns of the multinomial results. Detailed comparisons and robustness checks for both the macro and compositional models are reported in [Appendix C](#).

5.2 Opening Popularity

Opening popularity was modeled primarily as a compositional problem. The main inferential candidate was a pooled multinomial model fit to month-by-time-control-by-rating-bin

cells, with openings aggregated into a retained top- k set plus an “Other” category after sparse cells were excluded. Because classical cells were much less consistently supported after this aggregation and sparse-cell filtering, the retained pooled specification was restricted to blitz and rapid. This restriction preserves the rating-by-time-control comparison in the two best-supported formats while avoiding a comparatively unstable classical contrast. The specification used rating bin, time control, month index, and rating-by-time-control interactions to model opening composition across player-strength strata and formats.

Per-opening negative binomial and weighted least squares models were also fit as lower-complexity alternatives. These models were useful as sensitivity checks because they allowed individual openings to be examined one at a time, but that same feature limited their suitability as the primary model for opening choice. Since they do not model the opening set jointly, they cannot represent substitution across openings in a fully coherent way.

The pooled multinomial model was retained as the primary specification for this subsection. The main reasons were structural appropriateness and competitive held-out performance. It aligned most directly with the compositional form of the outcome while remaining interpretable enough for downstream use in Section 6. The simpler one-opening alternatives remained useful as supporting checks, but not as the main retained specification.

Robustness checks did not materially alter that decision. Allowing modest nonlinear flexibility in rating changed held-out fit only slightly, while greater flexibility in time did not improve performance enough to justify additional complexity. Varying the number of explicitly modeled openings produced the expected tradeoff between simpler and harder prediction problems, but did not change the broader retention logic. Full model compar-

Figure 7: Retained pooled multinomial opening-popularity model.

$$\begin{aligned}\Pr(Y_i = k) &= \frac{\exp(\eta_{i,k})}{\sum_{j=1}^K \exp(\eta_{i,j})}, \quad k = 1, \dots, K, \\ \eta_{i,k} &= \alpha_k + \beta_{1k} \text{RatingBin}_i + \beta_{2k} \mathbf{1}\{c_i = \text{Rapid}\} \\ &\quad + \beta_{3k} t_i + \beta_{4k} \text{RatingBin}_i \mathbf{1}\{c_i = \text{Rapid}\}, \quad k = 1, \dots, K-1, \\ \eta_{i,K} &= 0.\end{aligned}$$

Here, Y_i denotes the opening category for month-by-time-control-by-rating-bin cell i , t_i is the month index, and k indexes the retained opening categories, including “Other.” Blitz is the reference time-control category; classical cells were not included in this pooled specification. Category K is the baseline opening category against which the other opening categories are compared.

isons and supporting diagnostics for this subsection are reported in [Appendix C](#).

5.3 Time Use

Time-use modeling was organized around two related targets— average time use and low-time frequency. Because these outcomes differ in scale and distribution, they were evaluated separately rather than forced into a single retained model family.

For average time use, candidate specifications were compared at both the macro month level and the player-month level. At each level, additive models were compared against interaction specifications that allowed the relationship between rating and time use to vary by time control. The interaction specifications outperformed the corresponding additive alternatives at both analytic levels, even if the numerical improvement was modest. Those gains were directionally consistent and supported the conclusion that a common rating slope across formats was too restrictive for mean time-use modeling. The interaction forms

were therefore retained as the primary models for average time use.

Figure 8: Retained player-level mean time-use model.

$$\begin{aligned} \text{AvgTime}_i = & \beta_0 + \beta_1 \text{Rating}_i + \beta_2 \mathbf{1}\{c_i = \text{Classical}\} + \beta_3 \mathbf{1}\{c_i = \text{Rapid}\} \\ & + \beta_4 \text{Rating}_i \mathbf{1}\{c_i = \text{Classical}\} + \beta_5 \text{Rating}_i \mathbf{1}\{c_i = \text{Rapid}\} + \varepsilon_i. \end{aligned}$$

Here, AvgTime_i denotes average time spent per move for observation i . Blitz is the reference time-control category, and the interaction terms allow the rating slope to vary by time control.

For low-time frequency, the main comparison was between a fractional quasibinomial specification and a binomial approximation. Since predictive performance did not clearly distinguish the two, model retention was decided on inferential suitability and outcome structure. The fractional quasibinomial was retained because it is better matched to a bounded proportion outcome observed at the cell level, especially when the outcome is concentrated near zero. The binomial approximation remains a useful sensitivity check, but not the primary retained model.

Figure 9: Retained fractional low-time model.

$$\text{logit}(\mathbb{E}[L_i \mid X_i]) = \beta_0 + \beta_1 \text{Rating}_i + \beta_2 \mathbf{1}\{c_i = \text{Classical}\} + \beta_3 \mathbf{1}\{c_i = \text{Rapid}\}.$$

Here, $L_i \in [0, 1]$ denotes low-time frequency for observation i , and X_i contains rating and time-control indicators. Blitz is the reference time-control category.

Robustness checks supported both retention decisions. Alternative functional forms produced only limited changes in held-out fit, suggesting that the retained models were not being selected solely because of a narrow specification choice. Detailed comparisons and diagnostics for the time-use models are reported in Appendix D.

5.4 Player Progression

Player-progression models were designed to assess whether month- t player behavior could be used to predict subsequent rating movement under a forward-in-time design. Three targets were retained for evaluation: 1-month rating change, 3-month rating change, and absolute 1-month rating change. The 3-month target is based on a smaller sample because it requires observing the same player several months later, whereas the 1-month targets preserve substantially more observations. This difference in available support is important context for later comparisons across targets.

The main inferential candidate in this subsection was a player fixed-effects model fit to the player-month panel. This choice followed directly from the substantive goal of the task— isolating within-player associations between current-month behavior and later rating movement. The baseline feature set included current rating, game volume, average moves, opening diversity, top-opening concentration, and an indicator for whether the player switched their most-played opening. Ridge regression and random forest models were also fit as predictive benchmarks.

Across the retained targets, the player fixed-effects specifications provided the strongest overall combination of longitudinal appropriateness, inferential usefulness, and out-of-sample performance. This made the fixed-effects family the clearest retained direction for the progression task. Among observations for which player fixed effects could be estimated, the panel-based specifications outperformed the predictive benchmarks.

Sensitivity checks further supported that choice. Adding collapsed time-use features on the common-support sample did not improve performance enough to justify the accompanying loss of coverage, and allowing rating to vary nonlinearly produced only a small

Figure 10: Generic form of the retained player fixed-effects progression model.

$$Y_{i,t}^{(h)} = \alpha_i + \beta_1 \text{AvgRating}_{i,t} + \beta_2 \text{Games}_{i,t} + \beta_3 \text{AvgMoves}_{i,t} \\ + \beta_4 \text{OpeningDiversity}_{i,t} + \beta_5 \text{TopOpeningShare}_{i,t} + \beta_6 \text{SwitchedOpening}_{i,t} + \varepsilon_{i,t}.$$

Here, α_i is a player fixed effect, and $Y_{i,t}^{(h)}$ denotes one of three retained progression targets:

$\Delta_1 \text{Rating}_{i,t}$, $\Delta_3 \text{Rating}_{i,t}$, or $|\Delta_1 \text{Rating}_{i,t}|$, where $\Delta_h \text{Rating}_{i,t} = \text{Rating}_{i,t+h} - \text{Rating}_{i,t}$.

change relative to the baseline fit. Additional checks also suggested that absorbing common month shocks through added month fixed effects did not improve fit enough to outweigh the cost in simplicity. Full predictive comparisons and supporting robustness checks for this subsection are reported in [Appendix E](#).

5.5 Player Categorization

The goal of the player-categorization task was not to identify latent player “types” in a strong inferential sense, but to retain a descriptive specification that could summarize broad behavioral structure in player-month-time-control observations. For that reason, model retention in this subsection emphasized sample coverage, cluster balance, stability, and interpretability rather than formal inferential criteria.

The candidate clustering sample provided complete coverage for repertoire features but only sparse overlap for stable time-use features in the joined design. This made time use unsuitable as a core clustering input in the retained specification. Instead, the retained feature set focused on standardized performance, outcome, repertoire, and activity variables, with time-use measures treated as potential downstream overlays rather than direct inputs to the clustering algorithm.

Candidate descriptive approaches included k -means clustering, hierarchical clustering, Gaussian mixture clustering, and dimensionality-reduction methods used in a supporting role. The retained specification used k -means with $k = 6$ on the standardized core-plus-repertoire feature set. This solution was preferred because it provided a reasonably balanced partition, used the best-supported merged sample, and remained straightforward to interpret as a descriptive segmentation.

Figure 11: Retained k -means clustering objective for player categorization.

$$\min_{C_1, \dots, C_6} \sum_{k=1}^6 \sum_{i \in C_k} \|x_i - \mu_k\|_2^2.$$

Here, $x_i \in \mathbb{R}^p$ denotes the standardized feature vector for observation i , C_k denotes cluster k , and μ_k denotes the centroid of cluster k . The retained solution uses six clusters.

The robustness and stability checks supported treating the retained clustering solution as useful, but only in a descriptive sense. Relative to the core-only alternative, the core-plus-repertoire feature set yielded a more balanced solution. Repeated k -means fits under the retained specification also converged to the same partition, indicating very strong initialization stability. At the same time, agreement with alternative clustering algorithms was only moderate, silhouette values were modest, and month-to-month agreement for overlapping players was limited. These diagnostics suggest that the retained segmentation captures real structure, but not sharply-separated or highly-persistent latent classes. Detailed specification, comparison, and stability diagnostics are reported in [Appendix F](#).

5.6 Summary

The retained models define a common analytic strategy across our five empirical domains. Weighted linear and multinomial models were retained for metagame dynamics depending on whether the target was macro or compositional; a pooled multinomial model was retained for opening popularity; interaction-based mean models and a fractional quasibinomial were retained for time use; player fixed-effects models were retained for progression; and a k-means clustering solution based on core-plus-repertoire features was retained for player categorization.

6 Results

This section presents the substantive findings from the retained models described in Section 5. The aim is no longer to compare candidate specifications, but to interpret the model outputs. Accordingly, the section emphasizes interpretable summaries of model output, including estimated effects, representative predicted values, and descriptive profile summaries where appropriate. Additional robustness checks and supplemental diagnostics are reported in Appendices C, D, E, and F.

6.1 Metagame Dynamics

The retained metagame models indicate that the opening ecosystem broadened over the study period, although the magnitude of that broadening differed across time controls. The clearest evidence concerns concentration and diversity. Across formats, opening use became less concentrated and the effective number of openings increased, which together

indicate gradual metagame diversification rather than convergence onto a narrower set of dominant choices.

Table 1 provides the main inferential summary for the retained macro models. In the baseline format, HHI concentration declined over time while the effective number of openings (N_{eff}) increased, and both trends were larger in rapid and classical than in blitz. These paired results are substantively aligned—declining concentration and rising effective diversity describe the same underlying shift toward a broader opening ecosystem. The interaction terms matter here because the rate of diversification was not uniform across formats. Relative to blitz, the broadening was more pronounced in rapid and strongest in classical.

Table 1: Estimated time-trend effects from the retained macro metagame models.

Outcome	Effect	Estimate	SE	<i>p</i> -Value
Draw Rate	Baseline Time Trend	-2.322×10^{-5}	1.849×10^{-5}	0.215
Draw Rate	Classical $\times$ Time	2.615×10^{-5}	0.0001073	0.808
Draw Rate	Rapid $\times$ Time	9.697×10^{-6}	1.909×10^{-5}	0.614
HHI Conc.	Baseline Time Trend	-2.283×10^{-6}	6.890×10^{-7}	0.002
HHI Conc.	Classical $\times$ Time	-1.694×10^{-5}	7.154×10^{-6}	0.022
HHI Conc.	Rapid $\times$ Time	-7.440×10^{-6}	1.502×10^{-6}	< 0.001
N_{eff}	Baseline Time Trend	0.1021	0.0310	0.002
N_{eff}	Classical $\times$ Time	0.5556	0.2485	0.030
N_{eff}	Rapid $\times$ Time	0.2411	0.0545	< 0.001
Average Moves	Baseline Time Trend	-0.0003957	0.0025	0.873

Continued...

Outcome	Effect	Estimate		SE	<i>p</i> -Value
Average Moves	Classical $\times$ Time	−0.01816		0.0153	0.240
Average Moves	Rapid $\times$ Time	−0.003840		0.0035	0.272
Upset Rate	Baseline Time Trend	9.540×10^{-5}	4.099×10^{-5}		0.024
Upset Rate	Classical $\times$ Time	0.0001011		0.0002997	0.737
Upset Rate	Rapid $\times$ Time	7.481×10^{-5}	6.067×10^{-5}		0.223

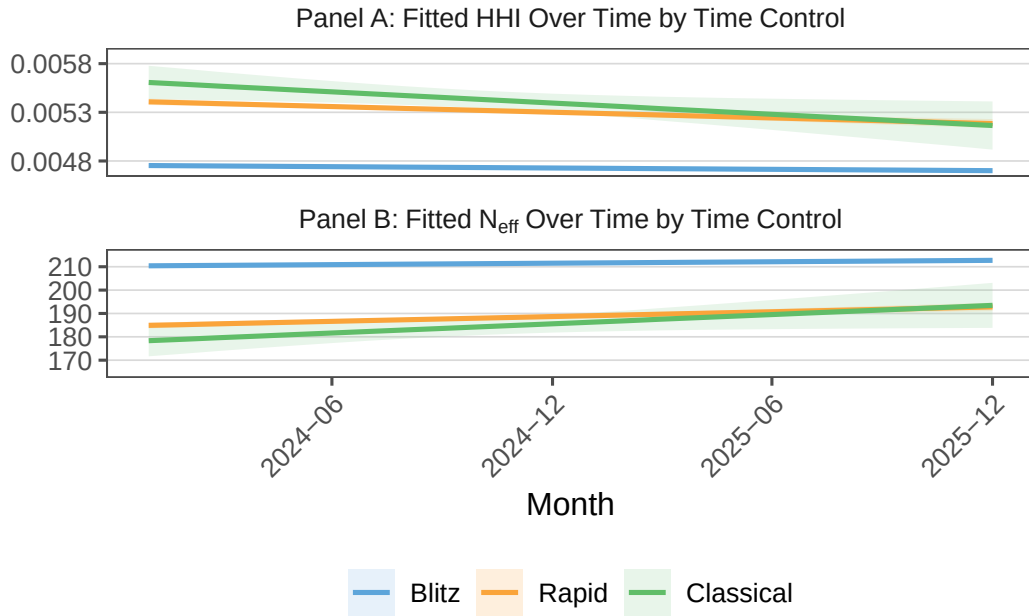
Note: Newey–West standard errors. Baseline time trend is for blitz; interaction terms are relative to blitz. “ N_{eff} ” represents number of effective openings.

Figure 12 makes the same pattern easier to see at the fitted-value level. The concentration curves trend downward over time, while the effective-diversity curves trend upward. The visual impression is not one of sudden disruption or rapid cycling, but of gradual structural widening across the two-year window. This is an important distinction: The metagame appears to be changing through accumulation and redistribution, *not* through abrupt replacement of one dominant opening by another.

By contrast, the retained macro models provide much weaker evidence for systematic time trends in draw rate and average moves. Those outcomes appear comparatively stable over the sample window. Upset rate shows some evidence of a modest positive baseline trend, but that signal is weaker and less consistent across time controls than the concentration and diversity results. This suggests that metagame structure changed more clearly than the broader game environment did.

Overall, the metagame results point to gradual diversification as opposed to convergence. The strongest findings concern ecosystem structure itself, where openings became less concentrated and more diverse, especially in slower formats. The compositional results

Figure 12: Fitted HHI concentration and effective number of openings over time by time control.



suggest that this broadening reflects diffuse redistribution across the opening ecosystem rather than a dramatic surge by any single named opening, with some of the increase appearing to occur in the long tail of less-common choices. Changes in draw rate, upset rate, and average moves were much less pronounced, so the main story here is structural broadening in opening choice. Additional metagame comparisons, compositional checks, and supplemental results are reported in Appendix C.

6.2 Opening Popularity

The retained opening-popularity model shows clear differences in opening choice across rating bins and time controls. The most visible pattern is the increasing share of the residual “Other” category as player strength rises.

Figure 13 provides the clearest visual summary of this pattern. In both blitz and

rapid, the predicted share of the “Other” category increases with rating.³ This implies that stronger players devote a greater proportion of their games to openings outside the explicitly retained top set. That result does not mean the named openings cease to matter; rather, it means the opening ecosystem becomes less concentrated around the same small group of common labeled choices as rating rises.

Figure 13: Predicted opening share by rating for selected openings in blitz and rapid.

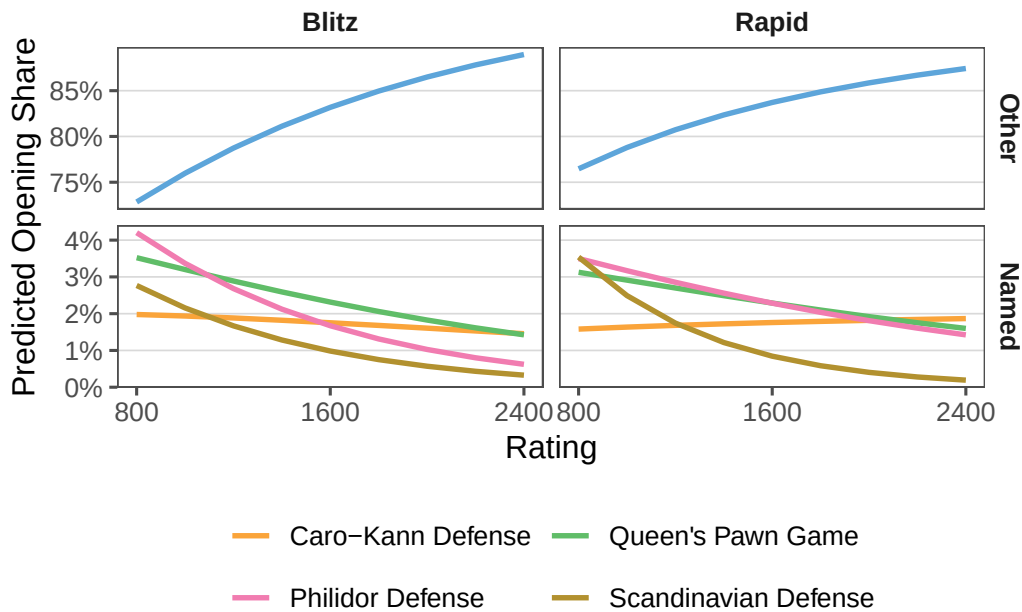


Table 2 provides representative predictions that help anchor the figure. At lower rating levels, more share is allocated to openings such as the Philidor Defense, Scandinavian Defense, and Queen’s Pawn Game. At higher rating levels, the top named openings shift toward alternatives such as the Caro-Kann Defense and the Accelerated London System, while the residual category becomes increasingly dominant. The exact ordering differs somewhat by format, but the broader skill gradient is consistent: Opening popularity is

³Because “Other” aggregates many lower-frequency openings, this pattern should be interpreted as reduced concentration in the retained top set rather than as a single coherent opening category gaining popularity.

strongly stratified by rating, and time control modifies that structure rather than replacing it.

Table 2: Representative predicted opening shares by rating and time control from the pooled multinomial model.

Rating	Time Control	Most-Likely	Second-Most	Third-Most
Bin		Opening	Likely	Likely
800	Blitz	Other (0.732)	Philidor Defense (0.041)	Queen's Pawn Game (0.034)
1600	Blitz	Other (0.835)	Queen's Pawn Game (0.023)	Queen's Pawn Game: Accelerated London System (0.017)
2400	Blitz	Other (0.892)	Caro-Kann Defense (0.014)	Queen's Pawn Game (0.014)
800	Rapid	Other (0.768)	Scandinavian Defense (0.034)	Philidor Defense (0.034)
1600	Rapid	Other (0.840)	Queen's Pawn Game (0.022)	Philidor Defense (0.022)

Continued...

Rating	Time Control	Most-Likely	Second-Most	Third-Most
Bin		Opening	Likely	Likely
2400	Rapid	Other (0.877)	Caro-Kann	Queen's
			Defense (0.018)	Pawn Game:
				Accelerated
				London System
				(0.016)

Temporal changes are present, but modest in magnitude. The over-time movement in the retained multinomial model is directionally consistent with the broader metagame results from Section 6.1: The residual “Other” category tends to gain share, while the largest named increases and decreases are comparatively small. This again points away from a story of dramatic named-opening turnover and toward one of distributed compositional change.

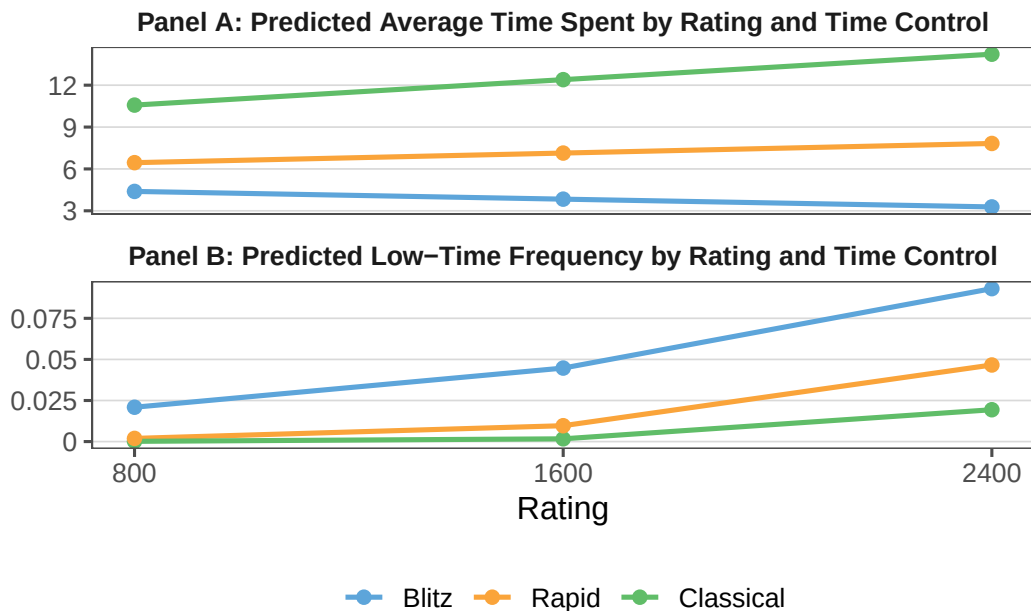
The substantive takeaway is that opening choice is highly structured, but most of that structure is cross-sectional as opposed to temporal. Rating and time control organize the opening landscape more strongly than month-to-month drift does. Over time, the opening ecosystem shifts, but the magnitude of that shift is much smaller than the difference between lower- and higher-rated opening distributions. Additional opening-popularity comparisons and supplemental results are reported in Appendix C.

6.3 Time Use

The retained time-use models show that average pacing and low-time pressure capture related but distinct aspects of chess behavior. Across the retained specifications, the most consistent result is that the relationship between rating and average time use differs by time control, while low-time frequency rises with rating in every format.

Figure 14 summarizes these two patterns. For average time spent per move, the rating gradient is negative in blitz but positive in rapid and classical. Higher-rated players therefore move faster on average in blitz, but spend more time per move in slower formats. This means that there is no single monotone relationship between skill and pacing. Instead, the behavioral meaning of a player’s use of time depends on the format in which it occurs.

Figure 14: Predicted average time spent per move and low-time frequency by rating and time control.



The size of the format differences is substantial. Blitz remains the fastest format across the rating range, classical the slowest, and rapid falls between them. More importantly,

the slope changes sign across formats, which supports treating time control as more than a nuisance adjustment; it changes the way player strength relates to deliberation time. A single common rating effect would have obscured this distinction.

The low-time results tell a different-yet-complementary story. Low-time frequency increases with rating in blitz, rapid, and classical alike, although the classical estimates are much less precise. The baseline prevalence of low-time states still differs sharply by format – with blitz much higher than rapid or classical – but within each format, stronger players are predicted to enter low-time situations more often. This is important because it shows that low-time frequency is not merely the inverse of average pace. A player can move faster on average and still enter acute time pressure more often, or spend more time on average and also enter low-time states more often.

These results suggest that time management is multi-dimensional; average pace reflects one aspect of behavior, but low-time exposure reflects another. In blitz, stronger players appear to play faster on average while also entering low-time states more often. In rapid and classical, stronger players spend more time on average yet still show higher low-time frequency. This combination implies that stronger players may be allocating time more selectively rather than simply playing uniformly faster or slower. Additional time-use diagnostics, robustness checks, and predicted profiles are reported in [Appendix D](#).

6.4 Player Progression

The retained player fixed-effects models relate within-player month-to-month behavior to later rating movement. Across the signed-change targets, the clearest positive predictors of later rating gain are average moves, opening diversity, and switching one’s most-played opening. By contrast, game volume and top-opening concentration show weaker and less

consistent evidence.

Table 3 provides the central inferential summary. For both the 1-month and 3-month signed targets, higher current average rating is associated with more negative subsequent signed rating change, while average moves, opening diversity, and switching the most-played opening are positively associated with later change. These effects are directionally consistent across the two signed targets, while games played and top-opening share show weaker and less consistent evidence.

Table 3: Estimated coefficients from the retained player fixed-effects progression models.

Predictor	Target	Estimate	95% CI	<i>p</i> -Value
Avg. Rating	1-Month Change	-0.469	[-0.511, -0.427]	< 0.001
Games Played	1-Month Change	-0.00295	[-0.0101, 0.00420]	0.398
Avg. Moves	1-Month Change	1.403	[1.213, 1.594]	< 0.001
Opening Diversity	1-Month Change	0.162	[0.129, 0.194]	< 0.001
Top-Opening Shr.	1-Month Change	6.697	[-3.782, 17.176]	0.196
Switched Opening	1-Month Change	2.722	[1.035, 4.409]	0.003
Avg. Rating	3-Month Change	-0.800	[-0.859, -0.740]	< 0.001
Games Played	3-Month Change	0.00345	[-0.00693, 0.0138]	0.494
Avg. Moves	3-Month Change	1.112	[0.870, 1.353]	< 0.001
Opening Diversity	3-Month Change	0.0614	[0.0165, 0.106]	0.010
Top-Opening Shr.	3-Month Change	-3.668	[-17.236, 9.901]	0.577
Switched Opening	3-Month Change	2.419	[0.249, 4.589]	0.031
Avg. Rating	Abs. 1-Month Change	0.00027	[-0.0107, 0.0112]	0.959

Continued...

Predictor	Target	Estimate	95% CI	<i>p</i> -Value
Games Played	Abs. 1-Month Change	0.00110	[-0.00469, 0.00689]	0.694
Avg. Moves	Abs. 1-Month Change	-0.334	[-0.503, -0.165]	< 0.001
Opening Diversity	Abs. 1-Month Change	-0.0712	[-0.0918, -0.0505]	< 0.001
Top-Opening Shr.	Abs. 1-Month Change	-5.837	[-13.635, 1.961]	0.133
Switched Opening	Abs. 1-Month Change	-3.755	[-4.344, -3.166]	< 0.001

Note: “Avg. Rating” represents current average rating; “Switched Opening” represents switching one’s most played opening.

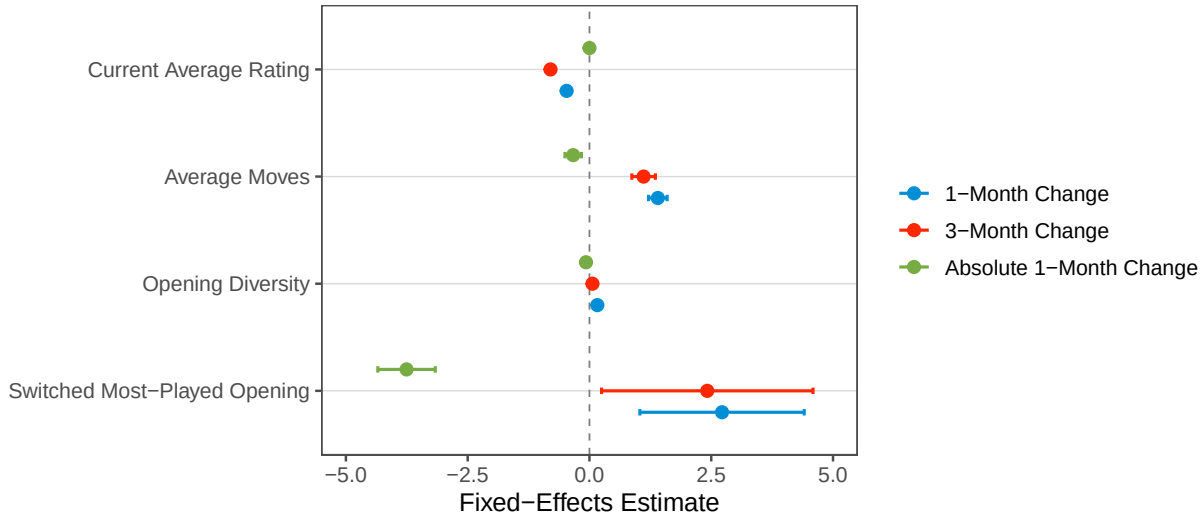
Figure 15 makes the main contrast clearer.⁴ Average moves, opening diversity, and switching one’s most-played opening are positively associated with later signed rating change, while current rating is negatively associated with it. The negative current-rating coefficient likely reflects short-run reversion and the mechanical difficulty of gaining additional rating from a higher current baseline, leaving the more behaviorally interesting signal in game complexity and repertoire adjustment.

The absolute-change model helps interpret those signed results. None of the favorable signed predictors is positively associated with greater movement magnitude; if anything, average moves, opening diversity, and switching are associated with smaller absolute month-to-month movement. This weakens the interpretation that the signed results are merely artifacts of noisier or more volatile play.

Substantively, the results point toward adaptation and complexity instead of volume alone. Months with longer games, broader repertoires, and a change in the most-played

⁴Coefficients are shown on their original predictor scales and are not directly comparable in magnitude across predictors.

Figure 15: Estimated coefficients from the retained player fixed-effects progression models by outcome target.



opening are associated with more favorable later signed rating movement, while raw game count is weak and inconclusive once these other behavioral measures are included. These results should still be interpreted as within-player associations, but they consistently suggest that richer and less static opening behavior is more closely linked to later rating gains than sheer activity volume. Additional predictive comparisons, fixed-effects robustness checks, and supplemental results are reported in Appendix E.

6.5 Player Categorization

The retained clustering results are descriptive rather than inferential, so the goal of this subsection is to summarize recurring behavioral profiles rather than to identify definitive player “types.” Even with that caution in place, the retained categorization solution reveals meaningful structure in the player-month data. The clusters differ along several behavioral axes at once, including rating, rating differential, game length, repertoire breadth, opening concentration, and opening switching.

Table 4: Performance and outcome means by cluster for the retained player-categorization solution.

Cluster	Share	Avg. Rating	Avg. Rating Diff.	Avg. Moves	Win Rate
1	0.128	1760.9	7.6	35.6	0.486
2	0.180	1214.6	-51.0	27.5	0.369
3	0.217	1518.4	-3.5	32.9	0.481
4	0.112	1649.3	2.2	37.6	0.427
5	0.162	1669.6	42.6	33.4	0.604
6	0.202	1506.1	-2.3	33.5	0.480

Table 5: Repertoire means by cluster for the retained player-categorization solution.

Cluster	Share	Opening Diversity	Top Opening Share	Switched Opening
1	0.128	114.8	0.088	0.443
2	0.180	16.6	0.166	0.619
3	0.217	48.1	0.095	1.000
4	0.112	17.8	0.154	0.712
5	0.162	16.3	0.153	0.727
6	0.202	44.1	0.169	0.006

Tables 4 and 5 summarize the core descriptive profiles. The clusters are not simply a relabeling of average rating. Instead, they combine performance, repertoire, and game-structure features into distinct configurations. Cluster 1 and Cluster 2 form a particularly clear contrast: Cluster 1 is associated with a broader repertoire, higher rating, and longer

games, whereas Cluster 2 is narrower, lower rated, and associated with shorter games. Cluster 3 is especially notable for very high switching, while Cluster 6 is more concentrated and stable in its opening behavior. Clusters 4 and 5 both include relatively stronger players, but Cluster 4 is marked by longer games and lower win rates, while Cluster 5 combines favorable rating differentials with high win rates and narrower repertoires. Collectively, these contrasts suggest that the retained solution is capturing multiple styles of engagement.

The cluster distributions also vary by time control. Some profiles are more common in blitz, while others appear more often in classical, with rapid tending to be more evenly distributed. These differences should not be over-read, especially where classical support is thinner, but they do suggest that format is part of the behavioral structure the clustering solution is capturing. The profiles are therefore not purely abstract player labels; they are partly shaped by the time-control context in which behavior is observed.

Figure 16: Month-to-month transition matrix for the retained player-categorization clusters.

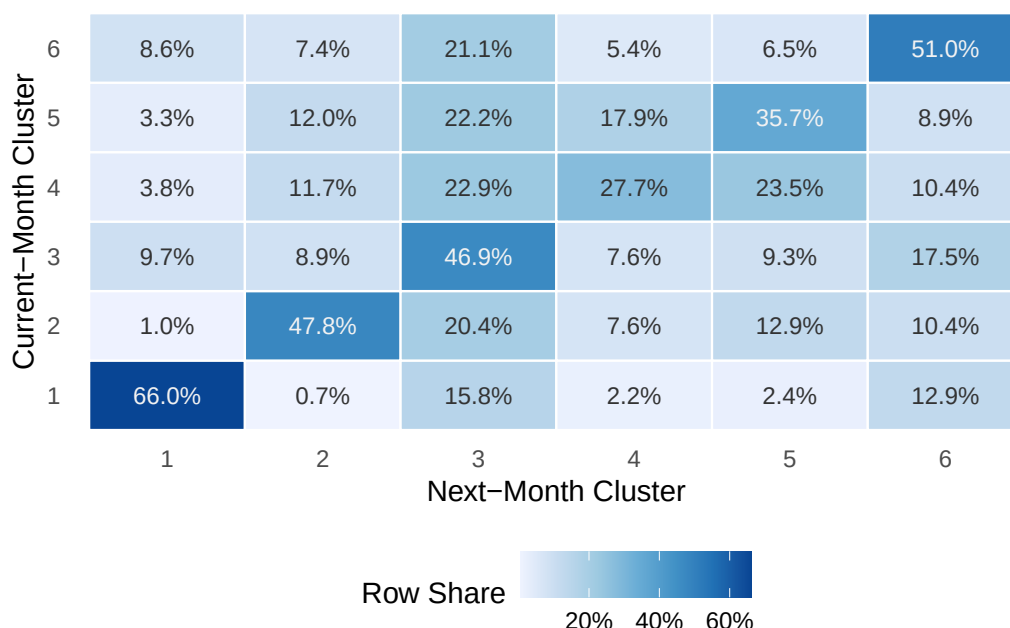


Figure 16 helps clarify how these profiles should be interpreted over time. Some clusters show substantial persistence from one month to the next, but switching is also common, and the transition structure is clearly patterned. This is the key reason the clusters are better understood as recurring behavioral states. They capture meaningful structure in player behavior, but that structure appears fluid enough that many players move among states over time instead of remaining locked into a single fixed profile.

This state-based interpretation is consistent with the rest of the paper’s results. The opening ecosystem broadens over time, opening choice varies with strength and format, time use is multi-dimensional, and progression is associated with adaptation. Against that background, recurring-but-fluid behavioral states are a more plausible interpretation than sharply-bounded and permanent classes. Additional clustering specification details, time-control composition summaries, persistence summaries, and robustness diagnostics are reported in Appendix F.

7 Discussion

7.1 Principal Findings

The retained models point to structured variation in online chess behavior across both ecosystem-level and player-level domains. At the ecosystem level, the metagame became less concentrated and more diverse over time. At the opening-choice level, popularity varied systematically by rating and time control. In the time-use domain, average pacing and low-time frequency behaved as related but distinct dimensions of player behavior. In the progression models, favorable later rating movement was associated more consistently with game complexity, repertoire breadth, and repertoire adjustment than with raw activity

volume. Finally, the clustering results suggested that player behavior is better summarized as a set of recurring behavioral profiles than as a small number of fixed player types.

In aggregate, these findings argue against simple, one-dimensional explanations. Metagame change was not primarily a story of one opening taking over. Time management was not reducible to a single measure of efficiency. Progression was not best explained by game volume alone. Instead, the results repeatedly point toward diversification, conditional behavior, and adaptation. The subsections that follow interpret those patterns in greater depth and consider what they imply about metagame change, player decision-making, development, and the broader study of online chess behavior.

7.2 Metagame Structure and Opening Choice

The strongest ecosystem-level finding was the paired decline in concentration and rise in effective diversity across time controls. In substantive terms, the metagame broadened over the study period. That result matters because a common expectation in competitive environments is some degree of convergence toward a relatively narrow set of dominant strategies, especially when players have broad access to shared theory, engine analysis, and educational content. The present results instead suggest a more expansive pattern, in which opening use became less concentrated rather than more so.

That conclusion should still be interpreted cautiously. Our results do not identify the mechanism behind this broadening, nor can they determine whether the same pattern would appear on other platforms or in other eras of online chess. Even so, the outcome is notable. One plausible interpretation is that large online player populations support a wider range of viable strategic habits than a narrower or more homogeneous competitive environment would. Another is simply that chess is strategically rich enough that increased

access to information does not mechanically force population behavior into a small set of dominant lines.

The opening-popularity results sharpen that picture. Opening choice varied much more clearly by rating and time control than by calendar time. Stronger players allocated less share to the most common named openings and more to the residual “Other” category, while time control modified that structure rather than replacing it. Temporal movement was directionally consistent with the broader metagame results, but comparatively modest in magnitude. The opening landscape therefore looks less like a rapidly-rotating cycle of named openings and more like a layered system of skill-stratified preferences with gradual long-run drift.

It is also useful to distinguish between two kinds of diversification. One possibility would be simple turnover among the same small set of highly popular openings. The other, which is more consistent with the present results, is a broader expansion in the share allocated outside that narrow top set. The latter interpretation is more substantively important, because it suggests widening participation in the opening space rather than mere reshuffling among a fixed group of choices.

Overall, these results support a picture of gradual diversification rather than convergence, with broadening driven more by long-tail expansion and skill-stratified choice than by visible turnover among a small set of named openings.

7.3 Time Management as Multi-Dimensional Behavior

The time-use results suggest that time management is not a one-dimensional behavioral trait. Average time use and low-time frequency are related, but they capture different

aspects of play. Across formats, the relationship between rating and average time use changed sign: Stronger players moved faster on average in blitz, but used more time on average in rapid and classical. At the same time, low-time frequency increased with rating in every format.

This matters because it changes the way time use should be interpreted. In blitz, faster average play among stronger players may reflect greater fluency, pattern recognition, or reliance on stable priors. In slower formats, greater average time use may instead reflect deeper calculation, more deliberate planning, or a higher willingness to invest time in critical positions. The same amount of time spent per move therefore does not have the same meaning across formats. Time control is not merely a nuisance adjustment; it changes the behavioral interpretation of pacing itself.

The low-time results reinforce that point. A simple expectation might be that stronger players use time more smoothly and therefore avoid acute time pressure. The retained models suggest otherwise. Stronger players were more likely to enter low-time states in blitz, rapid, and classical alike. This does not imply poor time management. Rather, it suggests that stronger players may allocate time more selectively, accepting later pressure in exchange for deeper thinking when they judge it worthwhile.

Together, these findings support a view of time management as multi-dimensional. Baseline pace and acute time-pressure exposure are not interchangeable, and neither can be reduced to a single global measure of “better” or “worse” time use. More broadly, the results suggest that time management is itself a meaningful behavioral domain rather than merely a mechanical byproduct of format.

7.4 Progression, Adaptation, and Player Development

A simple expectation about skill development is that more practice should lead to more improvement. In the context of online chess, that expectation would imply that raw activity volume is the strongest predictor of subsequent rating gains. The retained progression models did not support that view. Instead, more favorable later rating movement was associated more consistently with longer games, greater opening diversity, and switching the most-played opening. By contrast, games played was comparatively weak and inconclusive once these other behavioral measures were included.

This pattern suggests that player development may depend more on the character of engagement than on volume alone. Broader repertoires and changes in the most-played opening are consistent with adaptation, experimentation, or strategic flexibility. Longer games may likewise reflect more complex or demanding play. None of these interpretations can be treated as causal, but together they support a developmental account in which richer engagement matters more than repetition by itself.

The absolute-change model further clarifies the signed-change results. The predictors associated with favorable signed rating movement were not associated with larger absolute swings. If anything, average moves, opening diversity, and switching were associated with smaller absolute month-to-month movement. That pattern weakens the interpretation that the signed results are merely artifacts of noisier or more volatile play. The progression findings therefore do not look like a story of high-risk experimentation producing bigger swings in either direction. They look more like a story of gradual improvement associated with adaptation and complexity.

This does not mean that playing more games is irrelevant. A more careful interpre-

tation is that raw volume has limited additional explanatory value once more behaviorally informative measures are taken into account. In that sense, the retained models suggest diminishing returns to volume alone and a stronger role for adaptive engagement. Improvement, at least in this setting, appears better characterized as an adaptive process than as a purely cumulative one.

7.5 Behavioral Profiles as Descriptive States

The clustering results are best interpreted descriptively. They do not justify strong inferential claims about latent player “types,” but they do indicate that player-month observations contain enough structure to be grouped into recurring behavioral profiles. Those profiles vary along multiple axes at once, including skill, repertoire breadth, opening concentration, switching behavior, game length, and outcome structure. That is important because it means the clustering solution is not merely reproducing the platform’s rating hierarchy under a different name.

The retained profiles are most naturally interpreted as states rather than fixed identities. Some persistence is clearly present, but so are patterned transitions between clusters. Players do not appear to remain locked into a single behavioral class over time. Instead, the evidence suggests movement among recurring configurations of behavior. That reading is consistent with the rest of the paper’s findings: The opening environment broadens over time, time use depends on context, and progression is associated with adaptation rather than rigidity. Against that background, recurring-but-fluid behavioral states are a more plausible interpretation than sharply-bounded and permanent player types.

This interpretation also helps explain why the clustering results are useful despite their inferential limits. Their value does not come from proving the existence of a small

number of true latent player categories. It comes from showing that player behavior is structured enough to support interpretable recurring profiles. That in turn suggests a promising direction for later work— modeling movement among behavioral states directly.

7.6 Practical Uses and Broader Implications

Methodologically, the project shows that linked models of metagame structure, opening choice, time use, progression, and behavioral profiling can be integrated within a single analytic framework. These domains are often studied separately, but the present results suggest that they are better understood as linked parts of a broader behavioral system.

The results also point to possible practical uses in coaching or player-development settings. The progression models suggest that broader and more adaptive engagement may be more informative than game volume alone. The time-use results suggest that average pace and low-time pressure should be treated as distinct aspects of behavior rather than collapsed into one summary metric. Together, these findings suggest that player-facing tools may be more useful when they track repertoire breadth, adjustment, and time-use patterns alongside more familiar performance measures.

More broadly, the paper shows that large-scale online game logs can support behavioral analysis without reducing chess to engine evaluation alone. Chess is especially well suited to this kind of work because it combines standardized rules, rich decision structure, repeated play, and a mature rating system. As a result, it offers a useful environment for studying adaptation, strategic diversity, and development in a naturally occurring competitive setting.

7.7 Limitations

The most important limitation of the project is that it is observational throughout. The results support associations, not causal claims. For example, opening diversity and repertoire adjustment are associated with more favorable later rating movement, but the models do not establish that increasing diversity or switching openings causes improvement. Likewise, the time-use results describe behavioral signatures associated with skill and format, not causal mechanisms of stronger play. Even the fixed-effects models, while stronger than purely cross-sectional comparisons, do not eliminate time-varying confounding.

A second limitation concerns data scope and analytic breadth. The project was intentionally designed to be reproducible and to address several substantive questions within one framework. That choice made the analysis broader, but it also constrained how deeply any one domain could be modeled. A more specialized study could devote more of the sample, feature design, and model structure to a single problem, such as move-level decision quality, richer opening taxonomy, or transition dynamics among behavioral profiles.

The models also compress a strategically complex environment into tractable summaries. That compression is unavoidable. Chess is too rich to model exhaustively at scale, so each analytic task relies on reduced representations of a much more detailed behavioral space. Some simplifications are especially visible in the opening models, which rely on a top- k structure with a substantively meaningful but still heterogeneous “Other” category, and in the use of relatively simple functional forms in several retained models. These choices are reasonable for the present project, but they still limit how finely the results can characterize behavior.

Finally, the findings are specific to the Lichess platform and to the January 2024

through December 2025 study window. They should therefore be interpreted as results about that setting, not as definitive descriptions of online chess as a whole. Other platforms, populations, or periods may display similar patterns, but they may also differ in meaningful ways.

7.8 Future Work

A natural next step is to apply the retained framework more deeply within specific domains. Especially promising directions include move-level decision-quality analysis, richer opening annotation, longer player histories, and explicit modeling of transitions among behavioral profiles. These extensions would preserve the present paper’s core structure while allowing much deeper treatment of individual behavioral domains.

Several specific empirical questions also follow directly from the present results. One is whether metagame broadening reflects genuine experimentation or primarily long-tail expansion in opening choice. Another is whether the higher low-time frequency observed among stronger players reflects sharper positions, more selective time investment, or some other feature of high-level play. A third is whether repertoire adjustment is a precursor to improvement or merely one correlate of broader behavioral change. These questions are not fully answerable within the current design, but they are clearly motivated by the retained findings.

A final priority is external validation. Because the results are tied to one platform and one time window, cross-platform and cross-period replication would be especially valuable. Replication on other platforms, or on earlier and later Lichess periods, would help distinguish patterns that are specific to this environment from those that are more general features of online chess. If the framework proves stable across settings, it could support

a broader comparative research program on strategic behavior and skill development in competitive online environments.

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Appendix

A Data Reference and Construction

A.1 Variable Definitions

Table A.1: Variable definitions for the player-month (pm) table.

Variable	Type	Description
player	categorical	Deterministic pseudonymous user ID (e.g., u_01234567abcdefgh). Identifies a player across months in longitudinal summaries.
month	categorical	Month of play as YYYY-MM. Primary time index for longitudinal analysis.
games	numeric (count)	Number of games played by the player in the month. Also a reliability/weighting signal.
wins	numeric (count)	Count of wins in the month.
losses	numeric (count)	Count of losses in the month.
draws	numeric (count)	Count of draws in the month.
win_rate	numeric (proportion)	Wins divided by games. Primary performance measure in player-level models.
draw_rate	numeric (proportion)	Draws divided by games. Useful because draw propensity varies by skill and style.
avg_rating	numeric	Average of the player's pre-game rating across games in the month. Primary skill proxy.
avg_opp_rating	numeric	Average opponent pre-game rating in the month. Contextualizes performance and approximates strength of schedule.

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Variable	Type	Description
avg_rating_diff	numeric	Average (player rating minus opponent rating) across games in the month. Summarizes typical matchup difficulty.
avg_moves	numeric	Average game length in moves. Proxy for style and game complexity.
opening_diversity	numeric (count)	Number of distinct openings used in the month. Proxy for repertoire breadth.
top_opening	categorical	Most frequently played opening during the month (ties broken deterministically). Summary of dominant repertoire choice.

Table A.2: Variable definitions for the player-month-by-time-control (**pm_tc**) table.

Variable	Type	Description
player	categorical	Deterministic pseudonymous user ID. Identifies a player across months and time controls.
month	categorical	Month of play as YYYY-MM. Primary time index for player-period analysis.
tc_bucket	categorical	Time control bucket: blitz , rapid , or classical . Stratifies behavior by constraint level.

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Variable	Type	Description
games	numeric (count)	Number of games played by the player in the month/time-control cell. Reliability/weighting signal.
wins	numeric (count)	Count of wins in the cell.
losses	numeric (count)	Count of losses in the cell.
draws	numeric (count)	Count of draws in the cell.
win_rate	numeric (proportion)	Wins divided by games in the month/time-control cell.
draw_rate	numeric (proportion)	Draws divided by games in the cell. Useful because draw prevalence varies by time control.
avg_rating	numeric	Average of the player's pre-game rating across games in the cell. Primary skill proxy within time control.
avg_opp_rating	numeric	Average opponent pre-game rating in the cell. Contextualizes performance within format.
avg_rating_diff	numeric	Average (player rating minus opponent rating) in the cell. Summarizes typical matchup difficulty.
avg_moves	numeric	Average game length in moves for the cell. Proxy for style/complexity within format.

Continued...

Variable	Type	Description
opening_diversity	numeric (count)	Number of distinct openings used in the cell. Proxy for repertoire breadth within format.
top_opening	categorical	Most frequently played opening in the cell (ties broken deterministically).

Table A.3: Variable definitions for the player-opening summary (`pom_tc`) table.

Variable	Type	Description
player	categorical	Deterministic pseudonymous user ID. Identifies the player whose repertoire entry is being summarized.
month	categorical	Month of play as YYYY-MM. Time index for repertoire analysis.
tc_bucket	categorical	Time control bucket: <code>blitz</code> , <code>rapid</code> , or <code>classical</code> .
eco	categorical	ECO opening code (e.g., <code>C41</code>). Standardized opening taxonomy.
opening	categorical	Human-readable opening family name. Main opening identifier for repertoire analysis.
games	numeric (count)	Number of games the player used the opening in the month/time-control cell. Frequency weight.

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Variable	Type	Description
<code>win_rate</code>	numeric (proportion)	Opening-specific win rate for the player in the cell. Used to study effectiveness conditional on opening.
<code>draw_rate</code>	numeric (proportion)	Opening-specific draw rate for the player in the cell. Helps characterize risk profile and style.
<code>avg_rating</code>	numeric	Average player rating in games where the opening was played. Enables comparison of opening choice across skill.
<code>avg_rating_diff</code>	numeric	Average rating difference in games where the opening was played. Captures whether openings are used in tougher or easier matchups.

Table A.4: Variable definitions for the player time-use summary (`ptm_tc`) table.

Variable	Type	Description
<code>player</code>	categorical	Deterministic pseudonymous user ID. Identifies the player in the time-use summary.
<code>month</code>	categorical	Month of play as YYYY-MM. Time index for player-period time-use analysis.
<code>tc_bucket</code>	categorical	Time control bucket: <code>blitz</code> , <code>rapid</code> , or <code>classical</code> .

Continued...

Variable	Type	Description
<code>move_rows</code>	numeric (count)	Number of sampled move observations contributing to the summary. Primary reliability signal.
<code>games</code>	numeric (count)	Number of distinct games contributing move records to the player-month-time-control cell.
<code>avg_time_spent</code>	numeric (seconds)	Mean time spent per move. Captures typical deliberation time.
<code>median_time_spent</code>	numeric (seconds)	Median time spent per move; robust to outliers and occasional long thinks.
<code>p90_time_spent</code>	numeric (seconds)	90th percentile of time spent per move. Measures the upper tail of thinking time and occasional deep calculation.
<code>avg_clock_after</code>	numeric (seconds)	Mean remaining clock time after moves. Proxy for time management and pacing.
<code>low_time_rate</code>	numeric (proportion)	Fraction of sampled moves flagged as low-time. Proxy for time-trouble frequency.
<code>avg_rating</code>	numeric	Average player rating across sampled move records in the cell. Allows stratification by skill.

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Variable	Type	Description
avg_rating_diff	numeric	Average rating difference across sampled move records in the cell. Controls for opponent-strength effects.
avg_time_opening	numeric (seconds)	Average time spent per move in the opening interval, defined as moves 1–10.
avg_time_middlegame	numeric (seconds)	Average time spent per move in the middlegame interval, defined as moves 11–25. May be missing for cells with no sampled moves in this range.
avg_time_endgame	numeric (seconds)	Average time spent per move in the endgame interval, defined as moves 26 and later. Often missing because many games do not reach this threshold.

Table A.5: Variable definitions for the metagame opening summary (`mom_tc`) table.

Variable	Type	Description
month	categorical	Month of play as YYYY-MM. Primary time index for metagame analysis.
tc_bucket	categorical	Time control bucket: <code>blitz</code> , <code>rapid</code> , or <code>classical</code> .

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Variable	Type	Description
<code>eco</code>	categorical	ECO opening code. Standardized opening taxonomy useful for grouping openings into families.
<code>opening</code>	categorical	Human-readable opening family name. Primary opening identifier in population-level summaries.
<code>games</code>	numeric (count)	Number of sampled games for the opening within the month/time-control cell. Precision/support indicator.
<code>share</code>	numeric (proportion)	Fraction of sampled games in the month/time-control cell using the opening. Core metagame-adoption measure.
<code>white_win_rate</code>	numeric (proportion)	Fraction of games in the cell that White wins. Used for opening-level success patterns and color asymmetries.
<code>black_win_rate</code>	numeric (proportion)	Fraction of games in the cell that Black wins. Complements <code>white_win_rate</code> .
<code>draw_rate</code>	numeric (proportion)	Fraction of games in the cell that end in a draw. Draw propensity varies by opening and time control.

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Variable	Type	Description
<code>upset_rate</code>	numeric (proportion)	Fraction of games in which the lower-rated player wins (pipeline definition). Proxy for volatility and competitiveness.
<code>avg_abs_rating_diff</code>	numeric	Mean absolute rating gap between players in the opening/month/time-control cell. Context for outcome rates.
<code>avg_moves</code>	numeric	Mean game length in moves for the opening/month/time-control cell. Proxy for complexity and style at the population level.

Table A.6: Variable definitions for the metagame month-level summary (`mm_tc`) table.

Variable	Type	Description
<code>month</code>	categorical	Month of play as YYYY-MM. Time index for broad metagame summaries.
<code>tc_bucket</code>	categorical	Time control bucket: <code>blitz</code> , <code>rapid</code> , or <code>classical</code> .
<code>games</code>	numeric (count)	Number of sampled games in the month/time-control cell. Precision/support indicator and denominator for aggregate rates.

Continued...

Variable	Type	Description
<code>avg_white_rating</code>	numeric	Mean pre-game rating of White across sampled games in the cell. Summarizes the average strength of White players in that slice of the metagame.
<code>avg_black_rating</code>	numeric	Mean pre-game rating of Black across sampled games in the cell. Complements <code>avg_white_rating</code> .
<code>avg_abs_rating_diff</code>	numeric	Mean absolute rating gap between players in the cell. Context for outcome rates and competitiveness.
<code>avg_moves</code>	numeric	Mean game length in moves for the cell. Proxy for overall complexity and style at the month/time-control level.
<code>white_win_rate</code>	numeric (proportion)	Fraction of games in the cell that White wins. Useful for broad color-balance and outcome summaries by format and month.
<code>black_win_rate</code>	numeric (proportion)	Fraction of games in the cell that Black wins. Complements <code>white_win_rate</code> .
<code>draw_rate</code>	numeric (proportion)	Fraction of games in the cell that end in a draw. Draw prevalence varies by time control and month.

Continued...

Variable	Type	Description
<code>upset_rate</code>	numeric (proportion)	Fraction of games in which the lower-rated player wins (pipeline definition). Proxy for volatility and competitive unpredictability.
<code>opening_distinct</code>	numeric (count)	Number of distinct openings observed in the month/time-control cell. Summary measure of metagame breadth or opening variety.

Table A.7: Variable definitions for the population time-use summary (`mvm_tc`) table.

Variable	Type	Description
<code>month</code>	categorical	Month of play as YYYY-MM. Time index for population-level time-use summaries.
<code>tc_bucket</code>	categorical	Time control bucket: <code>blitz</code> , <code>rapid</code> , or <code>classical</code> .
<code>move_rows</code>	numeric (count)	Number of sampled move observations contributing to the month/time-control summary. Primary support indicator.
<code>games</code>	numeric (count)	Number of distinct games contributing move records to the cell. Useful for distinguishing move-level and game-level support.

Continued...

Variable	Type	Description
avg_time_spent	numeric (seconds)	Mean time spent per move across all sampled moves in the cell. Captures overall deliberation time at the population level.
median_time_spent	numeric (seconds)	Median time spent per move in the cell. Robust to outliers and long individual thinks.
p90_time_spent	numeric (seconds)	90th percentile of time spent per move in the cell. Measures the upper tail of thinking time.
avg_clock_after	numeric (seconds)	Mean remaining clock time after moves in the cell. Proxy for overall pacing and time-management style by month and format.
low_time_rate	numeric (proportion)	Fraction of sampled moves in the cell flagged as low-time. Population-level measure of time-trouble frequency.

B Exploratory Diagnostics and Support

B.1 Reliability

Figure B.1: Distribution of games per player-month in the sampled panel.

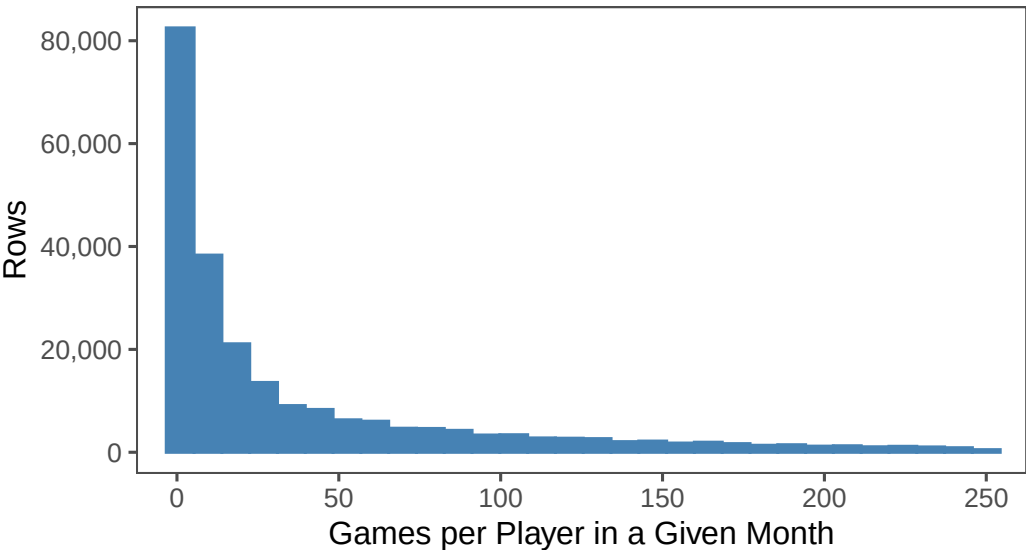
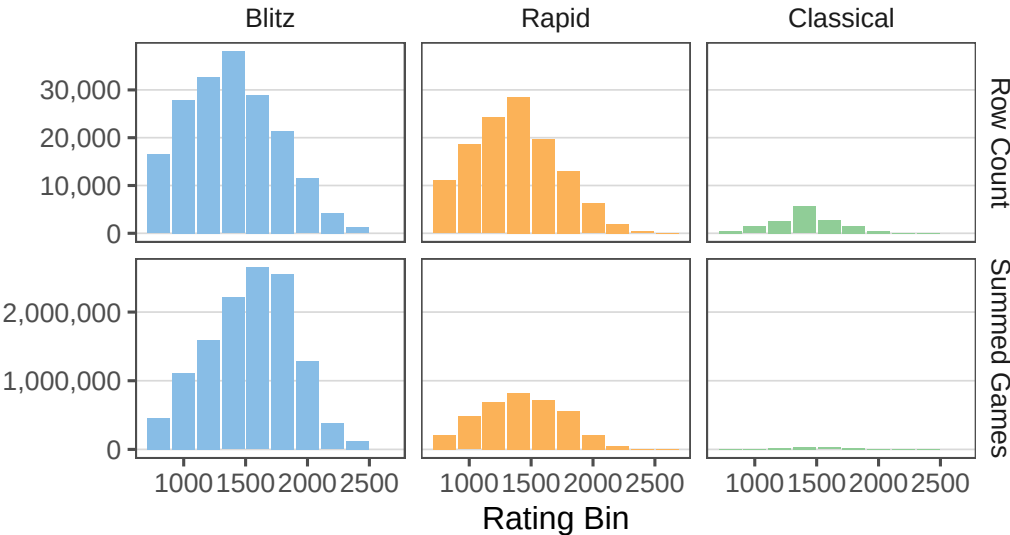


Figure B.2: Support across rating bins by time control, shown as row counts and summed games in `pom_tc`.



B.2 Rating-Change Noise Diagnostics

Figure B.3: Distribution of within-player month-to-month rating change.

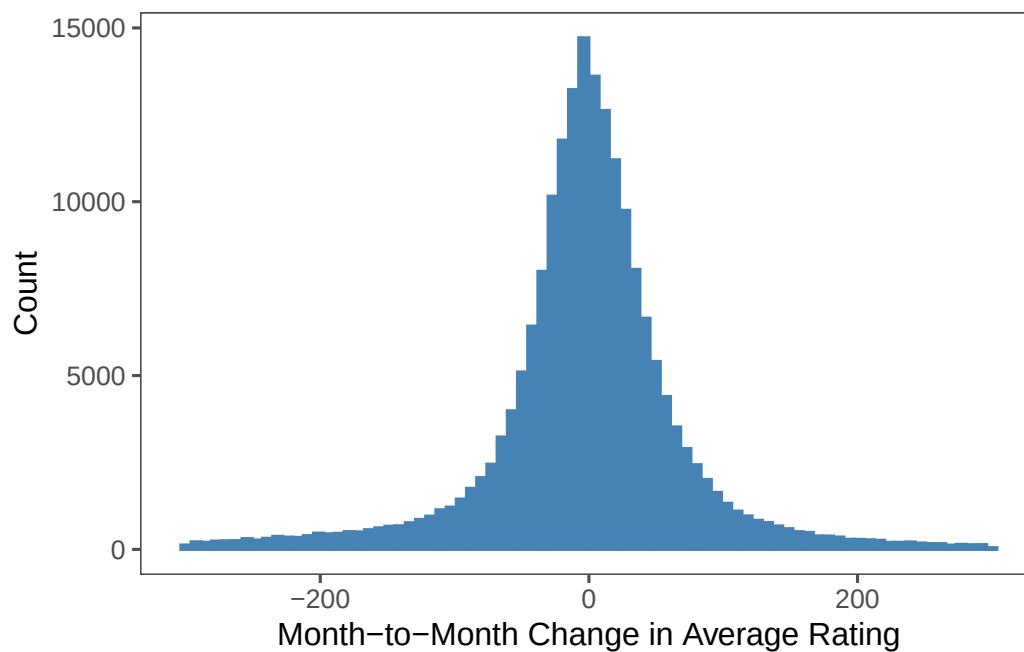
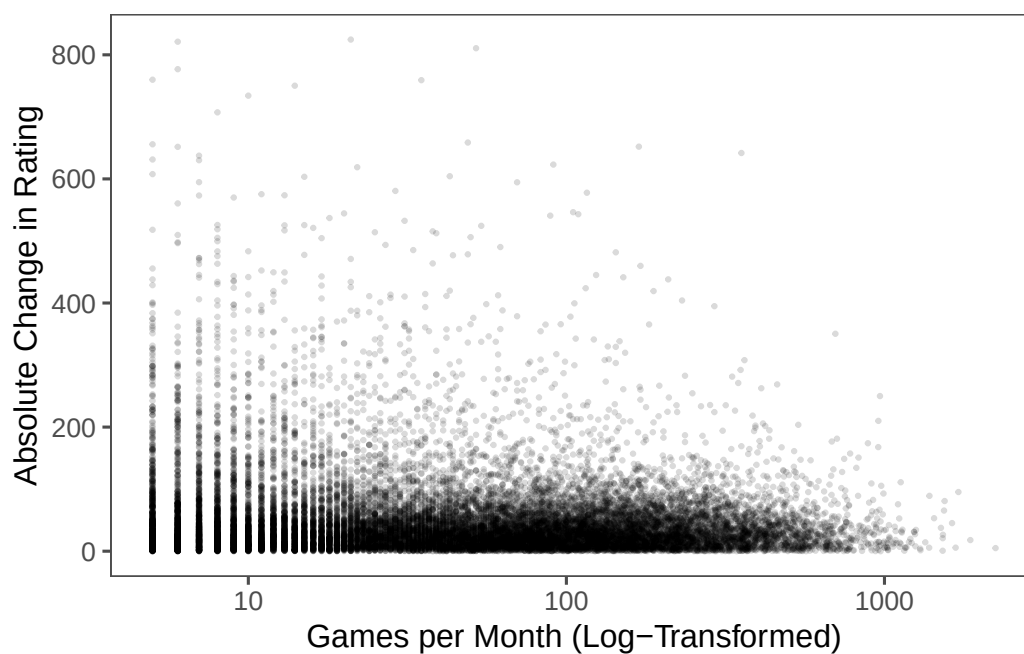


Figure B.4: Absolute month-to-month rating change by games played per month.



B.3 Time-Pressure Diagnostics

Figure B.5: Distribution of low-time rate and exact-zero mass by time control.

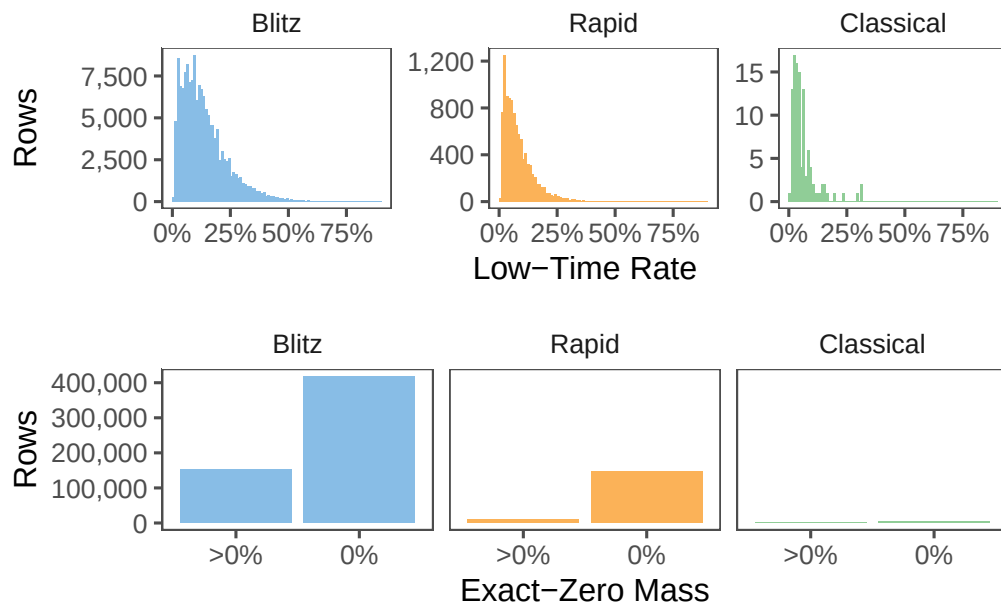
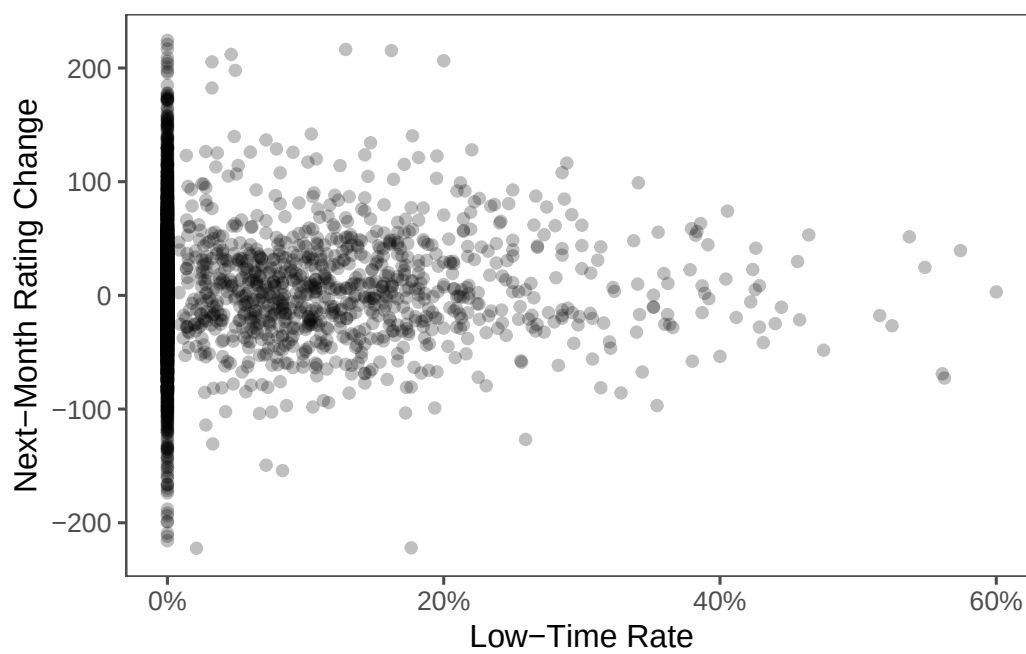


Figure B.6: Next-month rating change by low-time rate, pooled across time controls.



B.4 Supplemental Exploratory Plots by Topic

Figure B.7: Opening share over time for the most common blitz openings.

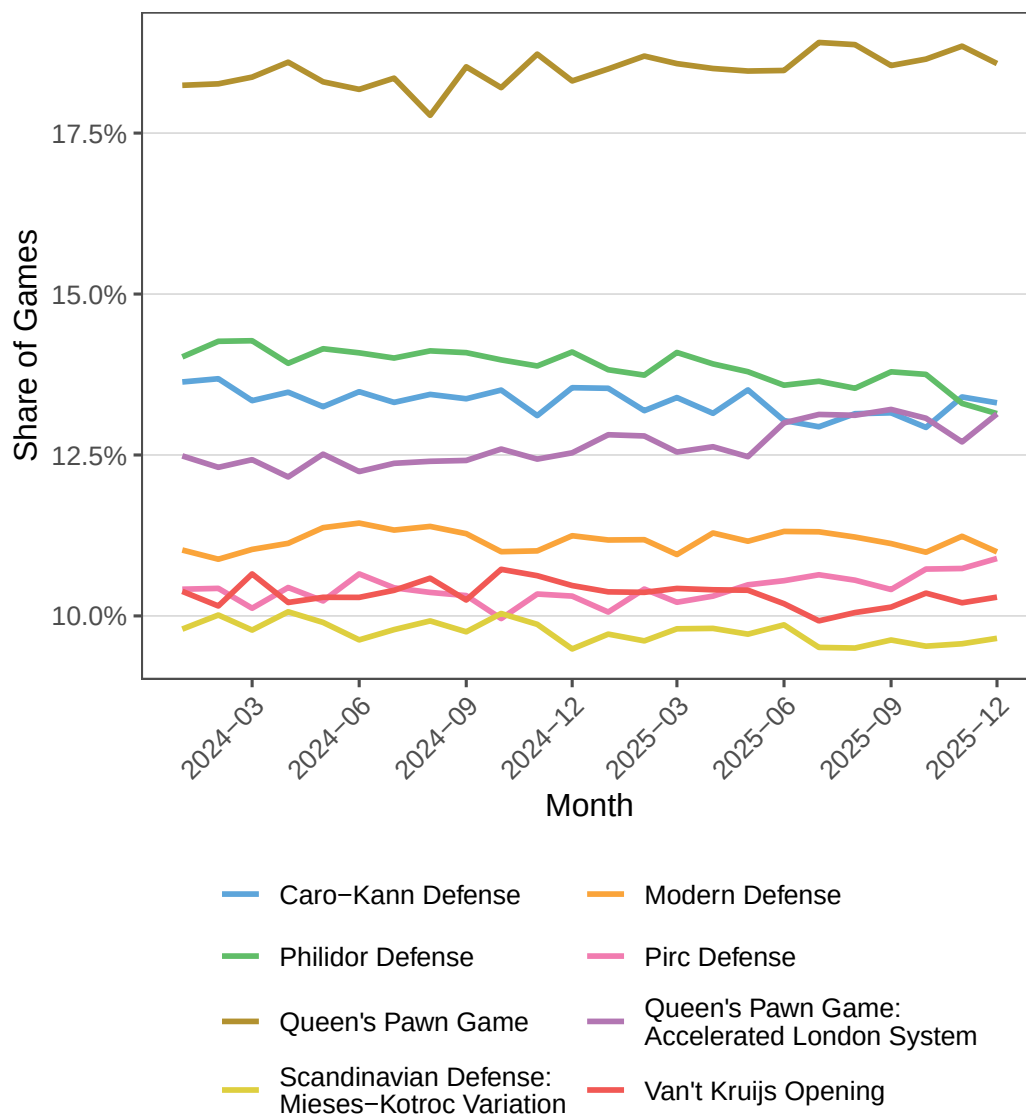


Figure B.8: Opening share over time for the most common rapid openings.

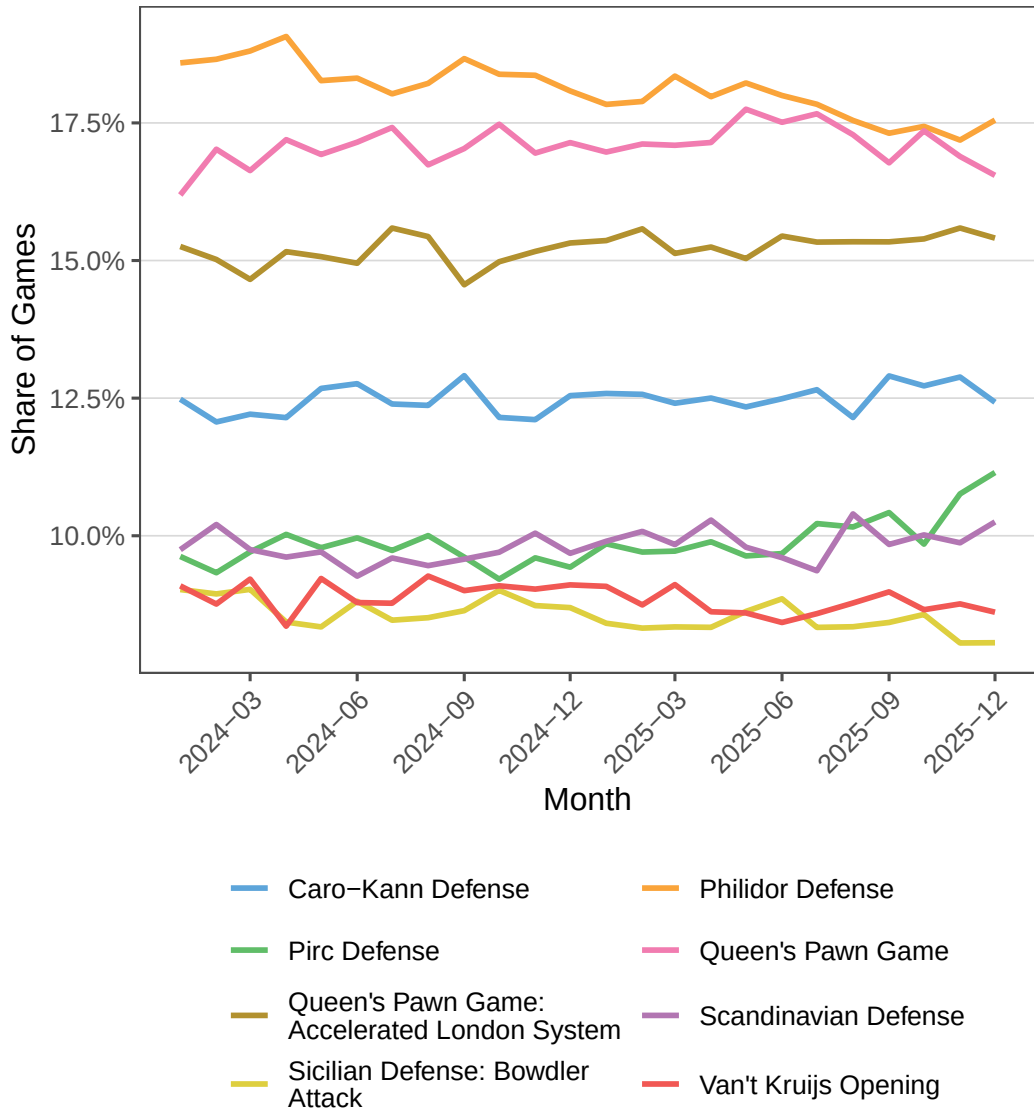


Figure B.9: Opening share over time for the most common classical openings.

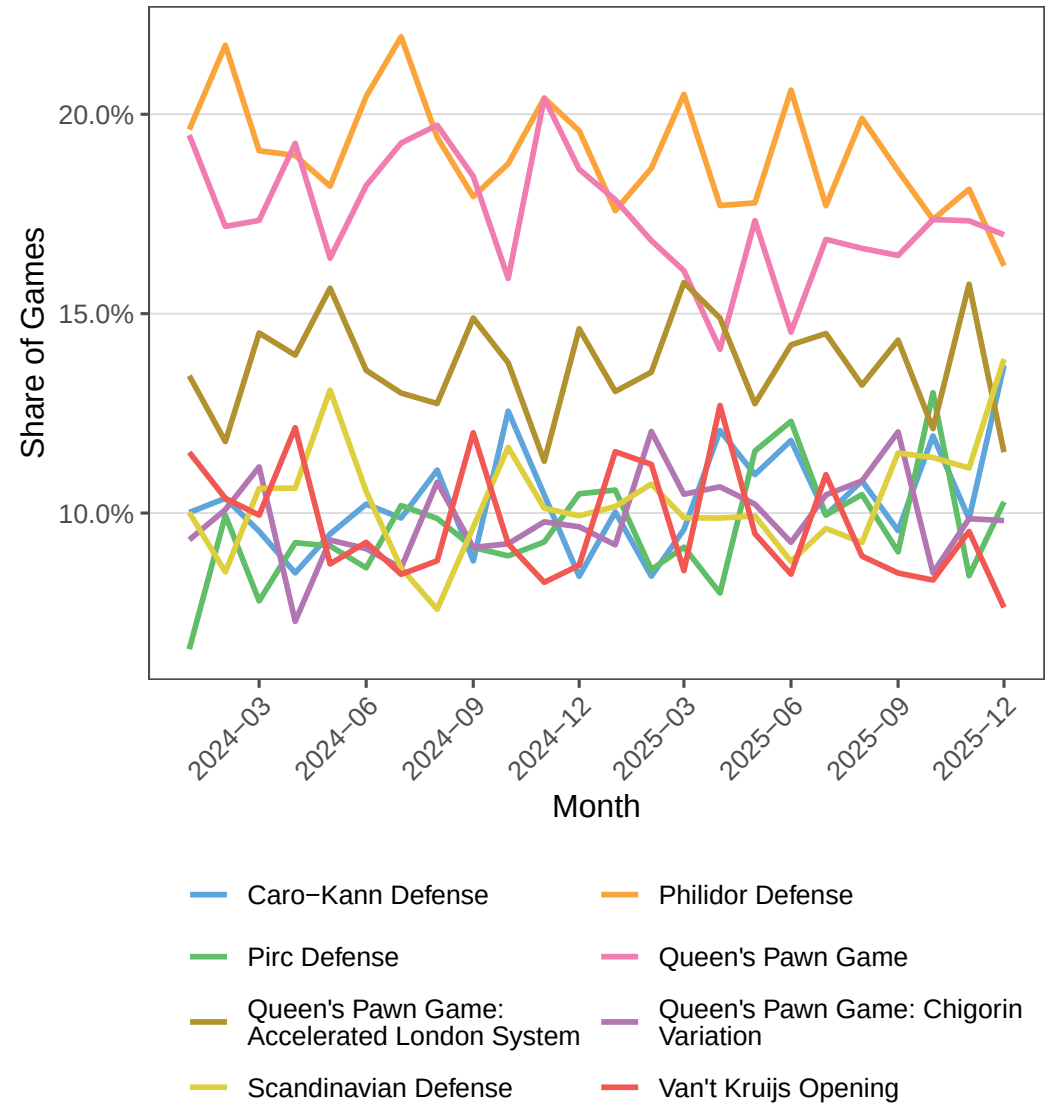


Figure B.10: Average time spent per move by rating, time control, and game phase.

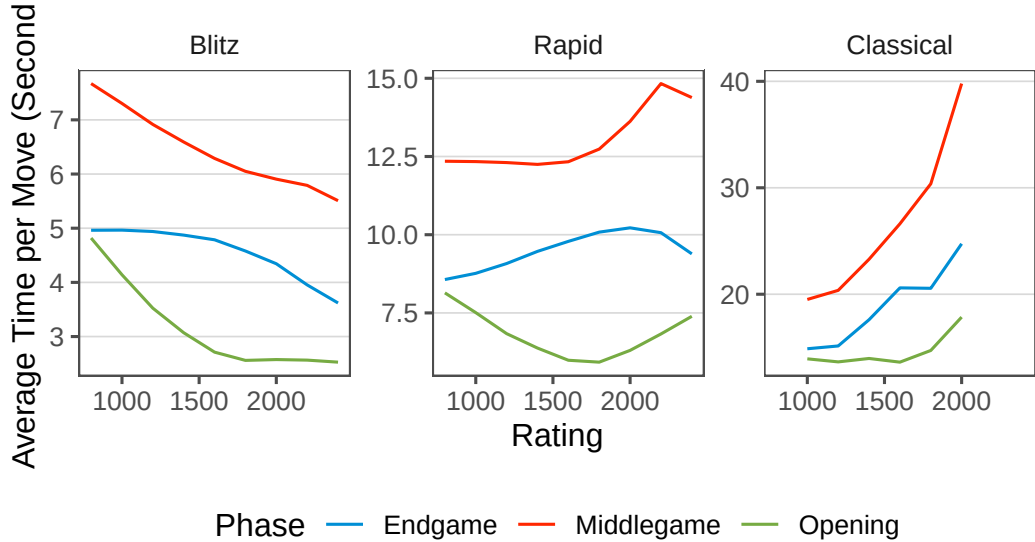
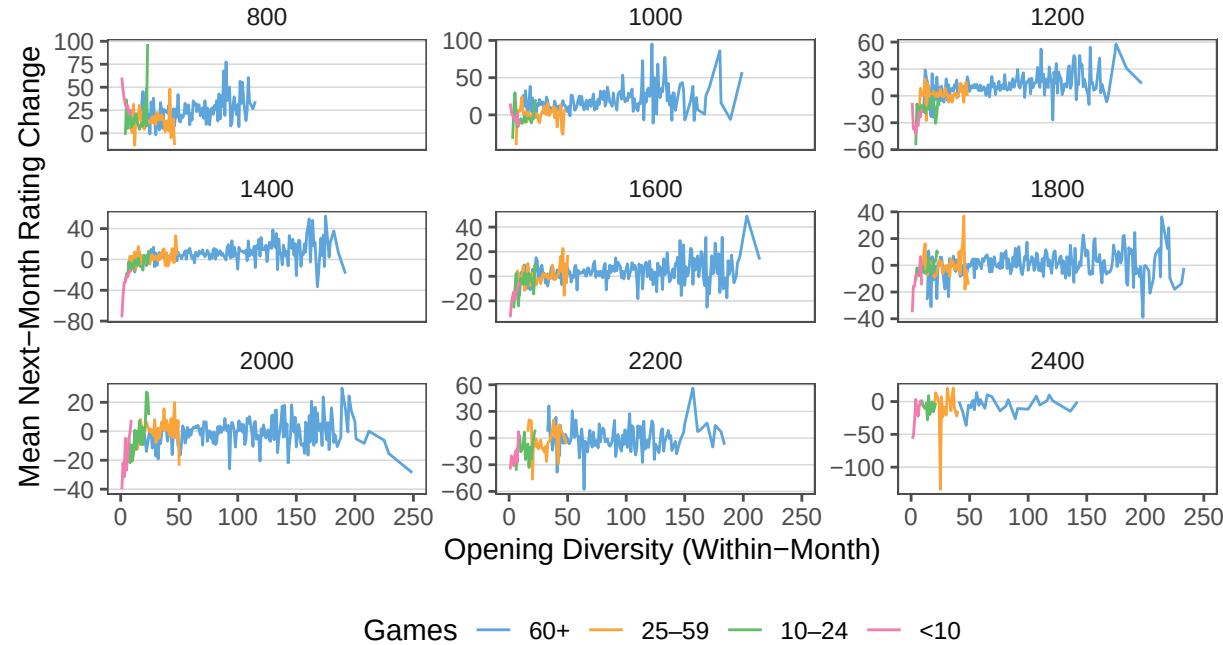


Figure B.11: Mean next-month rating change by opening diversity, stratified by rating and game volume.



C Metagame and Opening-Popularity Supplements

C.1 Metagame Model Comparison

Table C.1: Held-out predictive error for the retained macro metagame models.

Outcome	Time Control	Held-Out Error
HHI Concentration	All Formats	3.70×10^{-5}
Effective Number of Openings (N_{eff})	All Formats	1.48
Draw Rate	All Formats	0.00056
Upset Rate	All Formats	0.00158
Average Moves	All Formats	0.081

Table C.2: Held-out predictive error for retained opening-composition models.

Target	Model	Time Control	Held-Out Metric
Opening Share	FEOLS	Blitz	0.00243
Opening Share	GAM	Blitz	0.00243
Opening Share	FEOLS	Rapid	0.00239
Opening Share	GAM	Rapid	0.00237
Opening Composition	Multinomial	Blitz	0.910
Opening Composition	Multinomial	Classical	0.940

Continued...

Target	Model	Time Control	Held-Out Metric
Opening Composition	Multinomial	Rapid	0.923

Note: Metrics are comparable within model target and family, but not directly across share-regression and multinomial specifications.

C.2 Metagame Robustness and Residuals

Figure C.1: Observed versus fitted values for the retained macro metagame models.

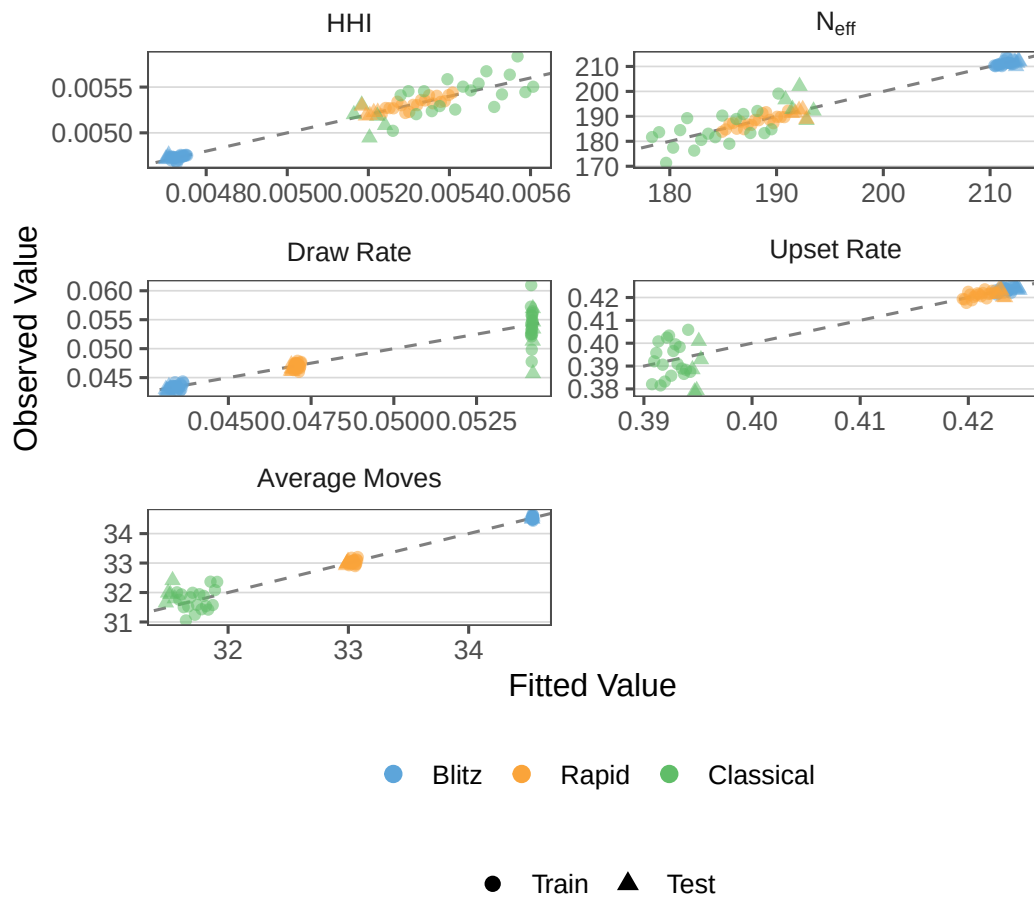


Table C.3: Robustness of the blitz multinomial model to the number of retained opening categories.

k	Categories	Log Loss	Top Increase	Top Decrease
8	9	0.674	Other (+0.000191)	Philidor Defense (−0.000162)
12	13	0.910	Other (+0.000951)	French Defense: Knight Variation (−0.000771)
16	17	1.127	Other (+0.001047)	French Defense: Knight Variation (−0.000771)

C.3 Opening-Popularity Model Comparison

Table C.4: Held-out predictive error for retained opening-popularity models.

Model	Target	Test Rows	Held-Out Metric
Pooled Multinomial	Opening Composition	81	0.893
Negative Binomial	Opening-Use Rate	80–81	0.0021–0.0097
Weighted Least Squares	Opening Share	80–81	0.0024–0.0046

Note: Metrics are comparable within model target and family, but not directly across share-regression and multinomial specifications.

C.4 Opening-Popularity Supplemental Results

Figure C.2: Observed versus predicted opening shares for the retained pooled multinomial model in the test set.

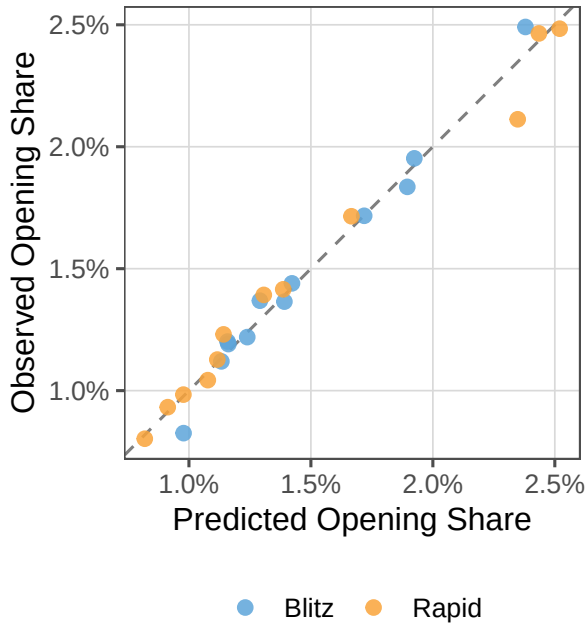


Table C.5: Largest predicted opening-share changes over time from the retained pooled multinomial model.

Time Control	Rating	Largest Positive Movers	Largest Negative Movers
Blitz	1600	Other (+0.0070); Queen's Pawn Game: Accelerated London System (+0.0023); Horwitz Defense (+0.0010)	French Defense: Knight Variation (−0.0041); Queen's Pawn Game (−0.0014); Pirc Defense (−0.0013)

Continued...

Time Control	Rating	Largest Positive Movers	Largest Negative Movers
Rapid	1600	Other (+0.0066); Queen's Pawn Game: Accelerated London System (+0.0029); Horwitz Defense (+0.0008)	French Defense: Knight Variation (−0.0038); Philidor Defense (−0.0014); Queen's Pawn Game (−0.0014)

Table C.6: Robustness checks for the retained pooled multinomial opening-popularity model.

Check	Specification	Log Loss
Rating Functional Form	Linear Rating	0.893
Rating Functional Form	Spline Rating	0.890
Time Functional Form	Linear Time	0.893
Time Functional Form	Spline Time	0.895
Top- k Threshold	$k = 8$	0.666
Top- k Threshold	$k = 12$	0.893
Top- k Threshold	$k = 16$	1.111

D Time-Use Supplements

D.1 Model Comparison

Table D.1: Held-out predictive error for retained mean time-use models.

Level	Specification	WRMSE
Macro Month-Level	Additive	0.149
Macro Month-Level	Interaction	0.141
Player-Month	Additive	1.839
Player-Month	Interaction	1.832

Table D.2: Held-out predictive error for retained low-time-rate models.

Specification	WRMSE
Fractional Quasibinomial	0.07445
Binomial Approximation	0.07445

D.2 Robustness and Residuals

Figure D.1: Observed versus predicted low-time rate for the retained fractional low-time model in the test set.

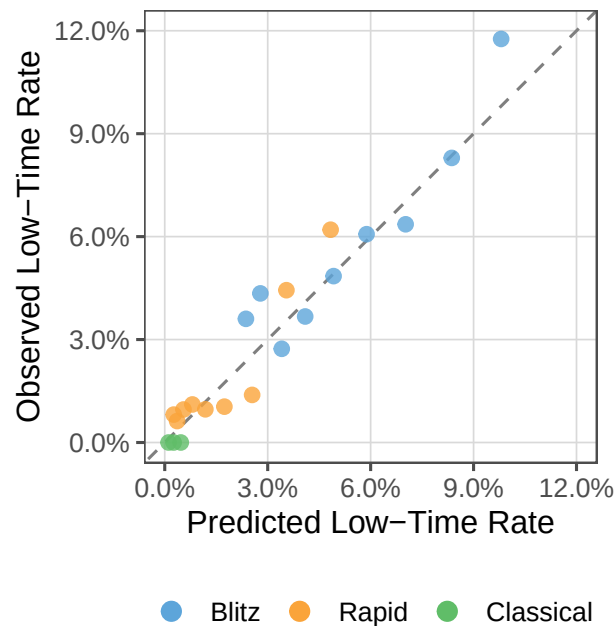


Table D.3: Robustness checks for retained mean time-use models.

Level	Specification	WRMSE
Macro Mean Time Use	Linear Time	0.141
Macro Mean Time Use	Spline Time	0.154
Player Mean Time Use	Linear Rating	1.832
Player Mean Time Use	Spline Rating	1.823

Table D.4: Robustness checks for the retained low-time-rate model.

Specification	WRMSE
Fractional Model, Linear Rating	0.07445
Fractional Model, Spline Rating	0.07445

D.3 Predicted Profile Tables

Table D.5: Predicted average time spent at the reference month from the retained player-level mean time-use model.

Time Control	Rating	Predicted Mean	95% CI
Blitz	800	3.973	[3.908, 4.038]
Blitz	1600	3.718	[3.677, 3.759]
Blitz	2400	3.463	[3.412, 3.514]
Classical	800	11.941	[9.083, 14.799]
Classical	1600	12.778	[11.150, 14.406]
Classical	2400	13.615	[11.172, 16.058]
Rapid	800	6.964	[6.791, 7.137]
Rapid	1600	7.280	[7.103, 7.456]
Rapid	2400	7.595	[7.286, 7.905]

Table D.6: Predicted low-time frequency at the reference month from the retained fractional time-pressure model.

Time Control	Rating	Predicted Rate	95% CI
Blitz	800	0.0209	[0.0183, 0.0239]
Blitz	1600	0.0447	[0.0414, 0.0483]
Blitz	2400	0.0931	[0.0856, 0.1011]
Classical	800	0.00014	$[1.28 \times 10^{-8}, 0.611]$
Classical	1600	0.00167	[0.000105, 0.0260]
Classical	2400	0.0194	$[6.54 \times 10^{-5}, 0.857]$
Rapid	800	0.00194	[0.00108, 0.00347]
Rapid	1600	0.00965	[0.00785, 0.0119]
Rapid	2400	0.0466	[0.0333, 0.0649]

E Player-Progression Supplements

E.1 Sample Construction and Predictive Comparisons

Table E.1: Sample sizes for player-progression targets after filtering and forward-month splitting.

Target	Train	Test	Definition
1-Month Change	99,049	18,166	$\Delta \text{Rating}_{t+1} = \text{Rating}_{t+1} - \text{Rating}_t$

Continued...

Target	Train	Test	Definition
3-Month Change	80,517	6,354	$\Delta \text{Rating}_{t+3} = \text{Rating}_{t+3} - \text{Rating}_t$
Absolute 1-Month Change	99,049	18,166	$ \Delta \text{Rating}_{t+1} $

Table E.2: Held-out predictive performance for retained player-progression models.

Model	Target	RMSE	MAE	Test Rows
Player Fixed Effects	1-Month Change	57.87	41.31	14,878 / 18,166
Ridge Regression	1-Month Change	61.62	41.31	18,166
Random Forest	1-Month Change	61.97	42.27	18,166
Player Fixed Effects	3-Month Change	70.76	50.83	5,726 / 6,354
Ridge Regression	3-Month Change	80.64	57.59	6,354
Random Forest	3-Month Change	81.38	58.42	6,354
Player Fixed Effects	Absolute 1-Month Change	41.74	28.56	14,878 / 18,166
Ridge Regression	Absolute 1-Month Change	45.67	30.38	18,166
Random Forest	Absolute 1-Month Change	45.70	30.66	18,166

Note: For fixed-effects models, test rows are reported as predicted / total because observations for players not present in the training panel cannot receive player fixed-effect predictions.

E.2 Fixed-Effects Robustness

Table E.3: Robustness checks for the retained 1-month fixed-effects progression model.

Specification	Train	Test Rows	RMSE	MAE
Baseline	99,049	14,878 / 18,166	57.87	41.31
Baseline, Common Support	2,015	62 / 405	27.86	21.23
Baseline + Time Use, Common Support	2,015	62 / 405	28.51	21.96
Linear Rating	99,049	14,878 / 18,166	57.87	41.31
Spline Rating	99,049	14,878 / 18,166	57.72	41.25

Note: Test rows represented as predicted / total.

Table E.4: Fixed-effects structure sensitivity for the retained 1-month progression model.

Specification	Train	Within- R^2
Player Fixed Effects	95,428	0.2696
Player + Month Fixed Effects	95,428	0.2744

E.3 Supplemental Interpretation

Figure E.1: Observed versus predicted mean 1-month rating change by decile of predicted change for the retained fixed-effects progression model.

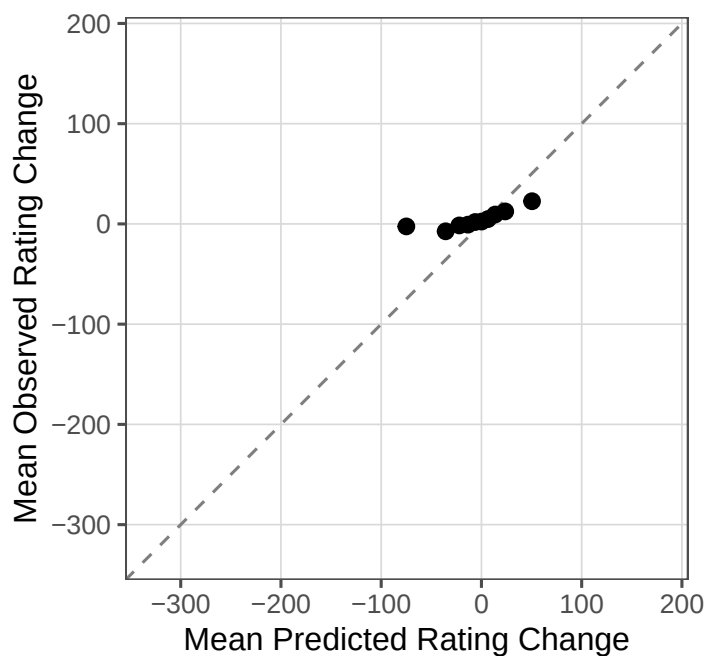


Table E.5: Representative predicted contrasts from the retained fixed-effects progression models.

Predictor	Target	Predicted Diff.	95% CI
Current Average Rating	1-Month Change	-247.000	[-267.706, -226.294]
Games Played	1-Month Change	-0.365	[-1.193, 0.462]
Average Moves	1-Month Change	9.668	[8.445, 10.891]
Opening Diversity	1-Month Change	7.434	[6.052, 8.816]
Top-Opening Share	1-Month Change	0.609	[-0.280, 1.498]

Continued...

Predictor	Target	Predicted Diff.	95% CI
Switched Opening	1-Month Change	2.722	[1.148, 4.296]
Current Average Rating	3-Month Change	-412.457	[-441.158, -383.755]
Games Played	3-Month Change	0.472	[-0.854, 1.799]
Average Moves	3-Month Change	7.443	[5.935, 8.950]
Opening Diversity	3-Month Change	2.883	[0.915, 4.852]
Top-Opening Share	3-Month Change	-0.324	[-1.441, 0.793]
Switched Opening	3-Month Change	2.419	[0.394, 4.443]
Current Average Rating	Absolute 1-Month Change	0.143	[-5.237, 5.523]
Games Played	Abs. 1-Month Change	0.136	[-0.533, 0.806]
Average Moves	Abs. 1-Month Change	-2.300	[-3.384, -1.215]
Opening Diversity	Abs. 1-Month Change	-3.274	[-4.161, -2.388]
Top-Opening Share	Abs. 1-Month Change	-0.531	[-1.192, 0.131]
Switched Opening	Abs. 1-Month Change	-3.755	[-4.305, -3.205]

Note: “Switched Opening” represents switching one’s most-played opening. The predicted difference for opening switching was calculated using values 0 and 1; predicted differences for all other predictors were calculated using the 25th and 75th percentiles.

F Player-Categorization Supplements

F.1 Specification and Cluster Comparison

Table F.1: Sample coverage and retained specification for player categorization.

Item	Value
Candidate Observations	156,901
Distinct Players	25,547
Distinct Months	24
Time-Control Levels	3
Fraction With Repertoire Features	1.000
Fraction With Time-Use Overlay	0.017
Retained Clustering Observations	156,901
Clustering Features	10
Primary Algorithm	K-Means
Chosen k	6
Feature Set	Core Plus Repertoire

Table F.2: Cluster sizes for the retained player-categorization solution.

Cluster	n	Share
1	20,107	0.128
2	28,181	0.180
3	34,057	0.217
4	17,546	0.112
5	25,390	0.162

Continued...

Cluster	n	Share
6	31,620	0.202

Table F.3: Feature construction for the retained clustering feature set.

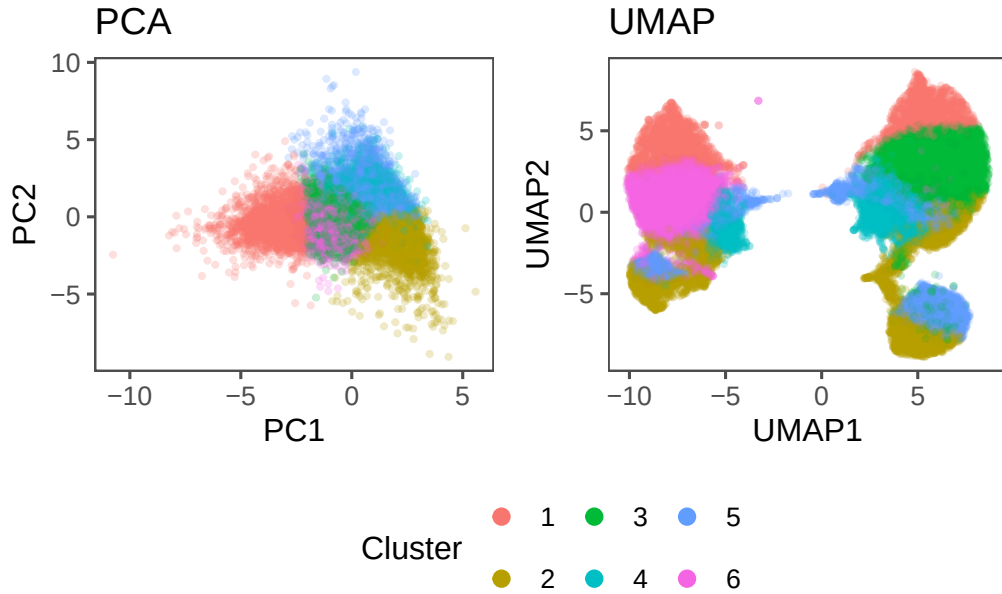
Feature	Source	Derived Construction	Transformation
avg_rating	pm_tc	Player-month-time-control age rating.	aver- None
avg_rating_diff	pm_tc	Player-month-time-control age rating differential.	aver- None
avg_moves	pm_tc	Player-month-time-control age moves.	aver- None
win_rate	pm_tc	Player-month-time-control rate.	win None
draw_rate	pm_tc	Player-month-time-control rate.	draw None
opening_diversity	pm_tc	Player-month-time-control opening-diversity already present in pm_tc.	summary None
top_opening_share_tc	pom_tc	max(games) / sum(games) within player $\times$ month $\times$ tc_ bucket.	None

Continued...

Feature	Source	Derived Construction	Transformation
switched_opening_tc	pom_tc	Indicator equal to 1 if the top opening differs from the previous month within player $\times$ tc_bucket, and 0 otherwise.	None
log_games	pm_tc	games from pm_tc.	log1p()
log_open_games	pom_tc	open_games_total = sum(games) within player $\times$ month $\times$ tc_bucket.	log1p()

Note: After construction and any feature-specific transformations, all retained variables were standardized as z-scores via `scale(as.matrix(X_raw))`.

Figure F.1: Two-dimensional PCA and UMAP projections of the retained clustering sample, colored by cluster.



F.2 Standardized Cluster-Center Views

Table F.4: Standardized performance and outcome centers for the retained player-categorization solution.

Cluster	n	Avg. Rating	Rating Diff.	Avg. Moves	Win Rate	Draw Rate
1	20,107	0.643	0.134	0.474	0.096	0.067
2	28,181	-0.888	-0.666	-1.001	-0.934	-0.423
3	34,057	-0.037	-0.016	-0.017	0.055	-0.110
4	17,546	0.330	0.061	0.826	-0.427	1.739
5	25,390	0.387	0.613	0.070	1.132	-0.525
6	31,620	-0.071	-0.000	0.095	0.041	-0.090

Note: Positive values indicate above-average cluster means on the standardized feature scale, while negative values indicate below-average means.

Table F.5: Standardized repertoire centers for the retained player-categorization solution.

Cluster	n	Op. Diversity	Top Share	Switched
1	20,107	1.994	-0.623	-0.286
2	28,181	-0.684	0.354	0.072
3	34,057	0.176	-0.534	0.845
4	17,546	-0.651	0.202	0.261
5	25,390	-0.690	0.191	0.291

Continued...

Cluster	n	Op. Diversity	Top Share	Switched
6	31,620	0.067	0.391	-1.171

F.3 Time-Control Composition and Persistence

Table F.6: Cluster shares within each time-control bucket.

Time Control	Cluster	Share Within Time Control
Blitz	1	0.164
Blitz	2	0.158
Blitz	3	0.227
Blitz	4	0.096
Blitz	5	0.140
Blitz	6	0.215
Classical	1	0.013
Classical	2	0.238
Classical	3	0.138
Classical	4	0.222
Classical	5	0.285
Classical	6	0.104
Rapid	1	0.064
Rapid	2	0.218
Rapid	3	0.202

Continued...

Time Control	Cluster	Share Within Time Control
Rapid	4	0.137
Rapid	5	0.199
Rapid	6	0.180

Note: Shares are computed within time control rather than across the full clustering sample.

Table F.7: Month-to-month persistence and most common transition destination for the retained player-categorization clusters.

Cluster	Adjacent Pairs	Same-Cluster Rate	Most Common Next Cluster
1	18,238	0.660	3 (0.464 of Switches)
2	12,599	0.478	3 (0.391 of Switches)
3	26,988	0.469	6 (0.329 of Switches)
4	8,924	0.277	5 (0.325 of Switches)
5	12,143	0.357	3 (0.345 of Switches)
6	25,661	0.510	3 (0.430 of Switches)

Note: Persistence is measured as the share of adjacent player-month pairs remaining in the same cluster.

F.4 Robustness

Table F.8: Robustness and stability diagnostics for the retained player-categorization solution.

Diagnostic	Statistic	Value
Feature-Set Robustness	Core-Only Min. Cluster Share	0.034
Feature-Set Robustness	Core Plus Repertoire Min. Cluster Share	0.112
Algorithm Agreement	K-Means vs. Hierarchical ARI	0.321
Algorithm Agreement	K-Means vs. GMM ARI	0.298
Silhouette	Mean Silhouette	0.145
Silhouette	Median Silhouette	0.151
Initialization Stability	Mean ARI	1.000
Initialization Stability	Min. ARI	1.000
Initialization Stability	Max. ARI	1.000
Month-to-Month Stability	Mean ARI	0.139
Month-to-Month Stability	Median ARI	0.143
Month-to-Month Stability	Min. ARI	0.025
Cluster Balance	Number of Clusters	6
Cluster Balance	Min. Cluster Share	0.112
Cluster Balance	Max. Cluster Share	0.217